

Critical Replication of Shaikh's Capacity Utilization Measure*

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Abstract

Capacity utilization depends on a productive ceiling that is never observed directly. [Shaikh \(2016\)](#) offers a portable accounting-based solution by deriving productive capacity from a long-run output–capital relation. This chapter critically replicates the coefficient that draws that capacity path, not only the utilization series built from it. Once reconstructed, the empirical output–capital coefficient is interpreted here as a candidate transformation elasticity, θ , linking capital accumulation to productive-capacity formation. Stage S0 finds that a Shaikh-like baseline is approximately recoverable from corporate national accounts, conditional on data-construction and specification choices. Stage S1 shows that a 500-model single-equation ARDL space does not collapse to a unique benchmark; some statistically retained alternatives also fail economic admissibility. Stage S2 moves the relation into a joint VECM system: the bivariate output–capital system is not self-sufficient, while a restricted trivariate system survives under the retained criteria when distribution and historical shock controls enter the long-run system. Shaikh's method is not discarded, but its structural interpretation requires explicit distributional and historical conditioning.

Keywords: Capacity Utilization; Critical Replication; Shaikh; Transformation Elasticity; Cointegration; ARDL; VECM; Distribution.

JEL Codes: B51; C32; E01; E22; E32; P16.

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1 Introduction

Capacity utilization begins from a measurement problem: actual output is visible, but the productive ceiling against which it is judged is not. [Shaikh \(2016\)](#) offers a portable accounting-based solution by constructing productive capacity from a long-run output–capital relation, rather than from survey utilization rates or official output-gap filters.

The narrow replication target is the coefficient that generates Shaikh’s productive-capacity path. Recovering the utilization series is not enough, because the series inherits its level, timing, and persistence from the long-run output–capital coefficient that determines the denominator of utilization.

The recovered empirical coefficient, d , then carries a stronger interpretation only after the theoretical mapping is stated. I read $\hat{d}$ as an estimated candidate transformation elasticity, $\hat{\theta}$, linked to the theoretical elasticity θ that connects capital accumulation to productive-capacity formation. That interpretive move is evaluated rather than assumed.

The empirical design follows the pressure points implied by that interpretation. Stage S0 reconstructs the closest recoverable Shaikh-like baseline; Stage S1 asks whether the coefficient is unique across the single-equation ARDL specification space; and Stage S2 tests whether the output–capital relation remains system-admissible when estimated in a joint VECM system.

Shaikh’s baseline capacity path is approximately recoverable, but the coefficient that generates it is not structurally self-sufficient. It is sensitive across statistically retained single-equation choices, and the bivariate output–capital relation fails the retained system-admissibility gates. Under the retained S2 criteria, a restricted trivariate system survives when the logged rate of exploitation and historical shock controls enter the long-run system. Some statistically retained estimates fail economic admissibility and are not treated as valid capacity benchmarks.

Shaikh’s approach is therefore narrowed rather than discarded. Its portability depends on making explicit the distributional and historical conditions under which the output–capital relation is stable. The chapter first locates the measurement problem historically, then defines the coefficient and its transformation-elasticity interpretation before turning to the three-stage replication and its bounded implications for profitability and capacity measurement.

2 A Historical Trace of Capacity Utilization Measurement

Capacity utilization entered macroeconomic measurement as a practical response to unobserved slack. In the postwar and Fordist policy setting, it translated idle plant and equipment into a language of demand management; after the inflationary crises of the 1970s, the same measure increasingly served as a signal of bottlenecks and inflationary pressure. Political economy later returned to the concept because neither survey utilization nor output-gap estimates resolved the deeper question of what determines the productive ceiling. The survey tradition is the starting point for that history.

2.1 From Survey Measures to an Inflation Sentinel: Capacity Utilization during the Fordist Era

Before governments could stabilize capacity utilization, they had to define and measure it. The Great Depression made idle resources a pressing political issue. Early researchers at the Brookings Institution tried to quantify this idle capacity, exposing a key tension: engineers defined capacity

as maximum physical output, while economists viewed it as the cost-effective level of operation. This distinction had major implications for policy (Burns, 1954; Klein and David, 1958). From the outset, utilization measurement operated as statecraft through numbers rather than neutral statistical description.

During the post-war era, Alvin Hansen’s secular stagnation thesis warned that chronic underutilization could derail long-run growth, forcing stabilization policy to confront the productive ceiling directly (Hansen, 1936, 1941; Barber, 1987). John Hicks and Paul Samuelson codified these concerns, embedding demand management into the IS-LM framework that structured U.S. graduate training (Hicks, 1937; Samuelson, 1947, 1948). Managing demand, employment, and bottlenecks required a metric that could translate idle capacity into intervention thresholds. Capacity utilization filled that role by rendering effective-demand shortfalls legible to a state apparatus committed to full-employment stabilization.

The McGraw-Hill plant-and-equipment surveys developed that role by standardizing how managers reported their actual and preferred operating rates. These surveys translated heterogeneous capacity definitions into a standardized reporting stream (Butler, 1958; Phillips, 1963). Institutionalizing the survey created a common language of economic slack for business forecasting, marginalizing deeper methodological debates over capacity measurement (Baran and Sweezy, 1988; Munroe, 2007). Later, the Federal Reserve integrated these survey-based capacity measures into its Industrial Production program, standardizing the benchmark as the default source for macroeconomic decision-making (Board of Governors of the Federal Reserve System, 2017). Alternative physical-intensity measures, such as the workweek of capital, never secured comparable bureaucratic adoption (Foss, 1963; Taubman and Gottschalk, 1971).

However, the stagflation crisis of the 1970s shifted the role of capacity utilization. Rather than signaling a need for expansionary policy to support employment, high utilization came to be viewed as a warning sign of upcoming inflation. Under the pressure of rising prices and falling profits, the metric was repurposed from an indicator of demand shortfalls to a sentinel for inflation control. Brookings debates increasingly judged capacity measures by their ability to deliver inflation forecasts, subordinating employment diagnostics to price stability (Klein et al., 1973). This durability did not resolve the underlying identification problem, but it cleared space for a new governance architecture based on output-gap benchmarks.

2.2 From Capacity Utilization to Output-Gap Governance

As global trade and financial networks expanded in the late 20th century, national manufacturing surveys became less central to policy. Central banks and international institutions needed portable, rule-based indicators to monitor economic activity across borders (Jessop, 2002, 2003; Konings, 2007). This demand led to the widespread adoption of the “output gap”—the difference between actual output and unobserved potential output. This shift was accompanied by a new emphasis on policy credibility, as formalized by the Lucas critique (Johnson, 2024). Discretionary demand management was replaced by technocratic, rule-bound frameworks like the Taylor rule, inflation targeting, and central bank independence (Taylor, 1993; Johnson, 2024). Saad-Filho (2018) identifies this settlement as the New Monetary Policy Consensus: a regime that depoliticized monetary policy and compressed wage shares to maintain price stability, driving the persistent rise in intergenerational income inequality observed since the 1980s (Hickel et al., 2026; Foster, 2024).

However, potential output is unobservable and must be estimated. Economists use various methods to construct it: statistical filters that smooth actual output trends (Hodrick and Prescott,

1997; Baxter and King, 1999; Beveridge and Nelson, 1981; Hamilton, 2018), production functions that combine labor and capital inputs (Congressional Budget Office, 2004; Havik et al., 2014; Alich, 2015), or structural vector autoregressions (VARs) that separate supply and demand shocks (Blanchard and Quah, 1989). These aggregate models also operate as a "black box" that hides the concrete workplace and market mechanisms of production. In reality, capital is converted into potential output (capacity) through concrete workplace processes. Purchasing equipment only establishes a theoretical capacity ceiling. Converting this capital into actual potential output depends on the intensity of labor and the organization of the shop floor. Class struggle directly mediates this conversion: labor resistance, wage bargaining, and supervisor discipline determine how intensively machines are run. Similarly, converting potential capacity into actual output is moderated by capital markets and firm expectations. This process is closely tied to the reserve army of labor and the unemployment rate: when unemployment is low, labor's bargaining power rises, squeezing profitability and prompting capital strikes that leave productive capacity idle, indicating that utilization is not a technical constant but a variable shaped by social conflict.

The Global Financial Crisis exposed the limitations of this output-gap framework. Post-crisis revisions routinely flipped the sign of the gap, and downward-biased potential-output estimates compressed public investment in the European periphery (Heimberger and Kapeller, 2017; Heimberger, 2020). While research divisions documented hysteresis and revision volatility (Cerra and Saxena, 2017; Cerra et al., 2023; Barkema et al., 2020), central banks accelerated the use of high-frequency nowcasting and forward-guidance models (Aastveit and Trovik, 2014; Berger et al., 2023; Foster, 2024). This revision-prone, closure-dependent apparatus remains entrenched in policy surveillance, providing the backdrop against which political economy's capacity-utilization debates must be understood.

2.3 Capacity-Utilization Debates in Political Economy

Within political economy, the debate over capacity utilization divides along a clear line: is the long-run normal rate of utilization endogenous to demand, or is it anchored in cost structures? The Neo-Kaleckian approach argues that the normal rate is endogenous. Firms adjust their desired operating rates in response to demand shocks, micro-level shift choices, or historical hysteresis (Lavoie, 1995; Nikiforos, 2013; Setterfield and David-Avritzer, 2020). In contrast, the Sraffian-Keynesian tradition argues that competition and cost minimization anchor the normal rate of utilization in the long run, with investment eventually adjusting capacity to meet demand (Ciccone, 1986).

However, both views face conceptual limits. The Sraffian supermultiplier framework requires a component of demand to remain permanently autonomous, which is difficult to reconcile with the debt dynamics of stock-flow consistent models (Nikiforos, 2018). Meanwhile, empirical tests remain inconclusive. While capacity utilization responds to growth in the short run, long-run convergence is hard to verify (Gahn and González, 2022).

Standard official indicators like the Federal Reserve capacity utilization series are ill-suited to resolve this debate. The Fed's series is smoothed by construction to reflect "normal" manufacturing operations rather than maximum technical capacity (Shapiro et al., 1989). It is also sectorally restricted to manufacturing, which represents a shrinking share of the modern economy (Haluska, 2023). Because the Fed series hardwires mean reversion, testing its stationarity merely reveals the Fed's own statistical smoothing protocols rather than actual economic gravity (Nikiforos, 2016, 2021b). Testing long-run capacity behavior requires a different empirical approach.

2.4 Shaikh’s Structural Identification of Capacity Utilization

To bypass these survey and smoothing constraints, Anwar Shaikh ([Shaikh, 2016](#)) shifts capacity estimation to profitability accounting. He uses a generalized perpetual inventory method to construct the capital stock and derives the capacity path from the accounting link between actual and maximum profit rates. In this framework, capacity utilization is defined as a residual deviation from the implied maximum output-to-capital ratio.

This accounting approach has the advantage of being portable, as it relies on standard national accounts and applies across different countries. However, treating capacity utilization as a deterministic residual assumes a stable relationship without testing whether the variables share a genuine cointegrating vector. If the underlying statistical relation is weak or misspecified, the resulting capacity utilization series will be highly unstable. This paper critically replicates Shaikh’s cointegration approach to stress-test its robustness and establish a firmer econometric foundation for measuring capacity utilization.

3 Conceptual Framework

The structural critique of aggregate production functions establishes that constant algebraic accounting identities must not be conflated with behavioral or structural laws of production ([Shaikh, 1974](#)). Although [Patomäki \(2017\)](#) outlines a plausible challenge to the closed-system ontology of [Shaikh \(2016\)](#) because it suppresses historical contingency, this critique targets philosophical appearance rather than analytical substance. The fragility in Shaikh’s account does not stem from mathematical abstraction itself, but from a failure of model closure that simultaneously omits distributive conflict and enforces an artificial algebraic identity. While omitting these institutional distributive dynamics constitutes an important specification vulnerability, resolving this omission remains outside the immediate scope of this essay. Instead, I focus strictly on this mismatch to show how Shaikh’s single-equation baseline substitutes a statistical determinacy for a genuine behavioral closure, forcing the aggregate long-run output–capital coefficient to operate as a time-invariant technical law rather than a historically contingent transformation elasticity of productive capacities ([Nikiforos, 2021a](#)).

This analytical limitation occurs because the framework fails to place the explanatory burden on the output–capital relation as a true transformation elasticity. Grounded in Marx’s schemas of reproduction, smooth proportional growth represents a restrictive analytical abstraction rather than a historical description of accumulation. Relaxing this balanced-growth baseline highlights the presence of systematic instability, where a behavioral assumption with deep theoretical underpinnings is required to govern the long-run development of productive capacity. Shaikh’s framework avoids this explicit structural parameterization, relying instead on the deterministic organization of an accounting identity. Consequently, the long-run coefficient remains a passive fitted slope that obscures the macroeconomic relation driving the transformation of accumulation into productive capacities.

Because the constructed capacity utilization series regulates the measurement of the normal rate of profit, this single-equation closure failure directly impacts the empirical findings of the classical surplus framework. Due to this direct functional dependency, the reconstructed utilization series inherits its dynamic properties from the foundational parameter, rendering subsequent conclusions regarding accumulation highly sensitive to the initial model choice. Instead of evaluating the statistical properties of the resulting residual, this section shifts the analytical focus toward the structural properties of this underlying coefficient. This section outlines these properties by isolating

the functional dependency of the unobserved series, examining how the accounting identity mimics a production law, and contrasting the macro-regimes of balanced versus unbalanced growth closures. Evaluating these conceptual boundaries requires a direct examination of the coefficient driving the productive-capacity path.

3.1 The coefficient driving the productive-capacity path

The empirical validity of a capacity-utilization series depends strictly on the stability of the parameter that generates the fitted capacity path. While [Shaikh \(2016\)](#) avoids direct physical proxies, his empirical output-capital relation can be grounded in a standard classical Leontief production function:

$$Y_t = \min\{A_t L_t, \mu_t B_t K_t\}, \quad (1)$$

where Y_t is aggregate output, L_t is employment, K_t is the capital stock, A_t is labor productivity, B_t is the maximum potential output-capital ratio at normal capacity utilization (denoted henceforth as the unobserved normal capacity-capital ratio, $R_t^n \equiv B_t$), and μ_t is capacity utilization. In a capital-constrained regime where labor is not binding due to the reserve army of labor, this production function simplifies to $Y_t = \mu_t B_t K_t$. Taking natural logarithms yields:¹

$$\ln Y_t = \ln K_t + \ln R_t^n + \ln \mu_t, \quad (2)$$

Within this specification, Y_t represents observed output, K_t represents the capital stock, R_t^n represents the unobserved normal capacity-capital ratio, and μ_t is capacity utilization. Both productive capacity $Y_t^p = R_t^n K_t$ and capacity utilization μ_t remain unobserved latent variables. While [Shaikh \(2016\)](#) utilizes a common price deflator to successfully eliminate inflationary distortions, this adjustment operates strictly within a monetary scale. Consequently, the specification retains a dimensional constraint because transcendental operations cannot carry constant-dollar units through the transformation, leaving the identity with uninterpretable unit structures like log-dollars ([Basu, 2022a](#)). Because logarithms evaluate underlying proportions rather than absolute levels, forcing unscaled monetary aggregates into the model renders the recovered parameters sensitive to the baseline unit of measurement.

Isolating the long-run capacity path requires modeling the unobserved normal capacity-capital ratio ($\ln R_t^n$) as a linear function of a deterministic time trend and the observed capital stock:

$$\ln R_t^n = bt + c \ln K_t. \quad (3)$$

Substituting this structural definition into the Leontief-derived log relation yields the estimable single-equation level relation:

$$y_t = a + bt + d k_t + \varepsilon_t, \quad d \equiv 1 + c, \quad (4)$$

Within this regression, y_t and k_t represent the natural logarithms of unscaled output and capital, while d represents the empirical long-run output-capital coefficient recovered from the level relation.

¹While [Shaikh \(2016\)](#) derives this relation strictly through the accounting identities of maximum profitability, we ground it here within the Leontief specification to provide a standard classical political economy representation of production that is immune to the Cambridge capital controversy. Because the Leontief framework assumes fixed technical proportions in the short run and discrete technique switching in the long run rather than smooth neoclassical marginal substitution, it provides a classical-safe foundation for the empirical output-capital level relation.

The constant a absorbs initial conditions and fixed measurement discrepancies in the levels of output and capital. The coefficient c governs how the normal capacity–capital ratio varies with capital, and the linear trend component (bt) is conventionally interpreted as a deterministic trend or autonomous growth component. If the underlying long-run behavior is instead driven by a proportional growth relation ($g_{Y^p} = \theta g_K$), this level trend functions as an algebraic adjustment factor that absorbs the scale differences between the unscaled time series.

The fitted path of productive capacity follows directly from the recovered long-run components:

$$\hat{y}_t^p = \hat{a} + \hat{b}t + \hat{d}k_t. \quad (5)$$

Capacity utilization is then constructed as the bounded deviation of realized output from this long-run capacity path:

$$\ln \hat{\mu}_t = y_t - \hat{y}_t^p, \quad \hat{\mu}_t = \frac{Y_t}{\hat{Y}_t^p}. \quad (6)$$

Because y_t and $\hat{y}_t^p$ share the same constant-dollar monetary units, this log-level subtraction algebraically collapses into the logarithm of a ratio. This operation allows the matching measurement units to cancel out completely, resolving the underlying scale mismatch and rendering the final extracted utilization series ($\hat{\mu}_t$) a dimensionally valid, unitless index (Basu, 2022a).

This estimation sequence establishes a direct functional dependency, where the constructed utilization series inherits its level, timing, and variance from the estimated parameter $\hat{d}$. Consequently, different single-equation specifications yield divergent $\hat{d}$ estimates, producing contradictory trajectories for both capacity and utilization. Because the model forces a dynamic accumulation process into a rigid accounting identity, the empirical sensitivity of this parameter raises an immediate identification problem. Structural measurement therefore rests on evaluating whether this long-run multiplier captures a genuine behavioral law of production, or whether it operates merely as an algebraic placeholder that absorbs omitted historical variation.

3.2 Laws of Algebra, Laws of Production, and Omitted Variable Bias

The estimated long-run coefficient d becomes a candidate transformation elasticity only after it is mapped onto the theoretical object $\theta \equiv \partial \ln Y^p / \partial \ln K$. Because the normal rate of profit relies directly on the constructed capacity–capital ratio, the structural integrity of the accumulation theory depends on whether this empirical coefficient can sustain that interpretation. While the aggregate accounting identity deterministically organizes these profit magnitudes, it leaves the underlying behavioral mechanism translating capital accumulation into productive capacity undefined. Following the analytical standard of Shaikh (1974), an accounting identity that leaves its core behavioral mechanism undefined mechanically mimics a structural production law, exposing the empirical framework to severe specification errors.

If the true long-run development of productive capacity operates as a proportional growth relation ($g_{Y^p} = \theta g_K$), the linear deterministic time trend inside the empirical specification functions as a mathematically superfluous level regressor. Simultaneously, the single-equation setup omits the historically shifting institutional vector, including distributive conflict and wage bargaining regimes. Omitting a relevant state vector while including an irrelevant deterministic trend generates a doubly misspecified model (Basu, 2020). Because the deterministic trend and the unobserved institutional shifts unfold contemporaneously across historical time, the trend acts as an accidental proxy. This correlation forces the estimated output–capital elasticity to absorb the omitted regime variation

through an indirect bias channel.

A constant capital stock yields highly divergent capacity trajectories depending on specific historical regimes, including shifts in labor militancy, wage bargaining, and financialization. Driven by the indirect bias channel, the estimated single-equation elasticity absorbs this institutional variance, appearing stable only because it mathematically averages across these structurally heterogeneous historical mappings. Formally evaluating whether this recovered parameter isolates a genuine structural relation requires explicitly defining the theoretical conditions of balanced and unbalanced accumulation.

Furthermore, we must address a key econometric limitation of estimating this transformation elasticity. The capital stock series K_t is constructed using the perpetual inventory method, which is notoriously subject to measurement errors regarding depreciation, asset pricing, and retirement patterns. Econometrically, measurement error in the right-hand-side variable (k_t) introduces a classical errors-in-variables problem. This error biases the estimated elasticity parameter θ (and the long-run coefficient d) downward toward zero. When interpreting the empirical estimates of θ , we must treat them as lower-bound estimates of the true transformation elasticity, acknowledging that capital stock measurement errors make the observed capacity response appear more sluggish than it is in reality.

3.3 Balanced and unbalanced growth

Marx’s schemas of reproduction provide the analytical benchmark for proportional macroeconomic growth (Okishio, 2022). Smooth capitalist reproduction requires strict material proportions: commodities must sell, surplus value must be realized, and the material composition of output must match the input requirements of continuous accumulation (Basu, 2022b). Because real capitalist economies routinely violate these proportions, smooth reproduction remains an analytical abstraction rather than a historical description. To isolate the supply-side relation between capital and capacity, the long-run classical closure ($\mu = 1$) holds demand instability constant (Foley, 1985). Canonical post-Keynesian models enforce a balanced-growth baseline by assuming a unitary transformation elasticity ($\theta = 1$) in the long run. Combined with demand fluctuations, this unitary assumption generates the characteristic properties of the Harrodian knife-edge.

Introducing an unbalanced growth closure ($\theta \neq 1$) achieves long-run identification by treating proportional accumulation as a restrictive special case. Relaxing the unitary assumption shifts the analytical focus away from the classical closure ($\mu = 1$). Under a non-unitary parameter, systematic instability is an inherent feature of capacity formation, distinct from temporary deviations driven by demand. The transformation elasticity (θ) governs the long-run development of productive capacity independently of short-run demand stabilization. In this setup, θ acts as a structural placeholder before its relation to distributive conflict is endogenized.

Defining the transformation elasticity as $\theta \equiv \partial \ln Y^p / \partial \ln K$, the analytical properties of the accumulation system depend on whether this parameter deviates from unity. Table 1 defines three macroeconomic regimes bounded by these conditions.

Table 1: Accumulation Regimes and the Productive-Capacity Relation

Regime	Parameter Condition	Long-Run Relation ($\hat{Y}^p, \hat{K}$)	System Tendency
Balanced Growth	$\theta = 1.0$	Capital and capacity expand proportionally	Knife-edge baseline
Overaccumulation	$\theta < 1.0$	Capital expands faster than productive capacity	Accumulation decelerates toward stagnation
Excess Capacity	$\theta > 1.0$	Productive capacity expands faster than capital	Accumulation accelerates without supply-side limits

Note: This table defines the theoretical regimes governing the system.

We note a key limitation of this classification scheme. This categorization of balanced growth, overaccumulation, and excess capacity is strictly bounded to a single-sector macroeconomic setting. In a multi-sector economy, overaccumulation and capacity dynamics are mediated by inter-sectoral competition, capital mobility, and structural demand shifts across different industries. By abstracting from these multi-sectoral allocations, the single-sector model isolates the aggregate capital-to-capacity relation, but it limits our ability to trace how sectoral imbalances spill over to affect the general rate of profit. This single-sector bounding should therefore be treated as an analytical limitation of the approach, serving as a simplified baseline for our empirical replication.

These regimes represent the analytical limits of the accumulation process. Under an unbalanced growth closure without autonomous technical change, the dynamic path of the net capital growth rate ($\hat{k} \equiv \dot{K}/K - \delta$) is governed by the following ordinary differential equation:

$$\frac{d\hat{k}}{dt} = (\theta - 1) \hat{k} (\hat{k} + \delta), \quad (7)$$

where δ represents the depreciation rate. For any positive net accumulation ($\hat{k} > 0$), the sign of $d\hat{k}/dt$ depends on the magnitude of $(\theta - 1)$. Net accumulation decelerates under $\theta < 1$, accelerates under $\theta > 1$, and remains stationary under $\theta = 1$. This mathematical instability reflects the accounting laws of capital accumulation under a non-unitary transformation elasticity, distinct from Harrodian demand-driven dynamics.

Formally, $d\hat{k}/dt$ is a capital acceleration term, representing the second derivative of the capital stock with respect to time. Because a perpetual acceleration or deceleration of capital growth is economically impossible over long horizons, this term must remain bounded near zero over the long run. In historical capitalist economies, this bounding is enforced by rate-limiting and temporary institutional factors. When accumulation accelerates too rapidly ($\theta > 1$), it hits bottlenecks like labor cost inflation (wage barriers), central bank interest rate hikes (monetary policy shifts), and financial leverage limits (capital market constraints). Conversely, when accumulation slows down ($\theta < 1$), state interventions, bankruptcy reorganizations, and cheapened capital assets eventually bound the stagnation tendency, preventing the acceleration term from diverging infinitely from zero.

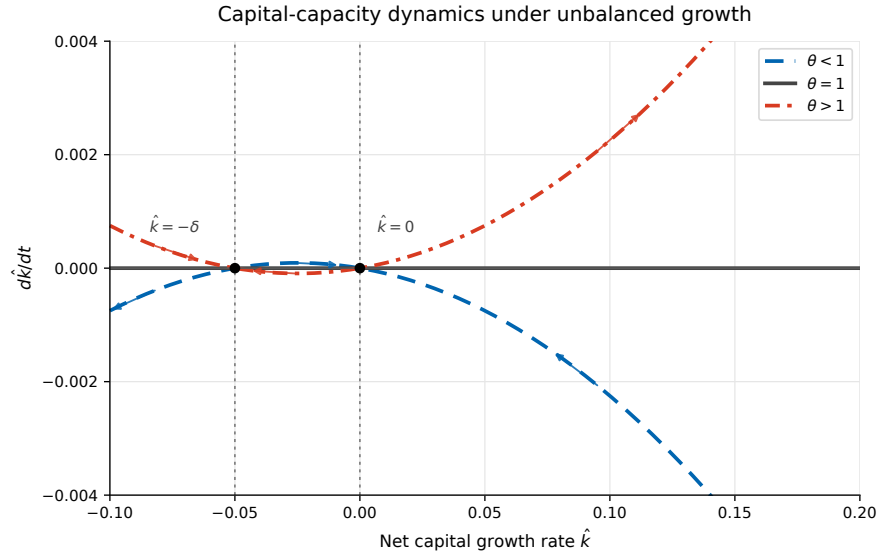
To illustrate this dynamic instability, consider a stylized numerical example of an overaccumulation regime. Let the baseline net capital growth rate be $\hat{k}_0 = 3\%$ (0.03) and the depreciation rate be $\delta = 5\%$ (0.05). If we assume a unitary transformation elasticity ($\theta = 1$), the growth rates of capital and capacity match exactly, and the net capital growth rate remains stationary

($d\hat{k}/dt = 0$). However, if the transformation elasticity is non-unitary, say $\theta = 0.8$, the system experiences overaccumulation. Substituting these values into equation (7) yields:

$$\frac{d\hat{k}}{dt} = (0.8 - 1) \times 0.03 \times (0.03 + 0.05) = -0.00048$$

This means the capital accumulation rate decreases by 0.048 percentage points per period. Economically, under an unbalanced growth parameter of $\theta = 0.8$, a 1% increase in capital accumulation relative to capacity formation entails a 0.2% capacity mismatch ($\theta - 1 = -0.2$), inducing a persistent downward drag on capital accumulation that decelerates the economy toward stagnation.

Figure 1: Capital accumulation dynamics under unbalanced growth



Note: The phase diagram plots $d\hat{k}/dt$ against $\hat{k}$ for $\theta < 1$ (dashed line), $\theta = 1$ (solid line), and $\theta > 1$ (dash-dot line). Vertical dotted lines mark the equilibria at $\hat{k} = 0$ and $\hat{k} = -\delta$.

While effective demand and distributive conflict retain their analytical relevance, they are abstracted from in this initial formulation. Institutional and macroeconomic forces dampen, accelerate, or redirect the underlying long-run tendency but do not generate it. The differential dynamics in equation (7) imply a regime-dependent trajectory, yet the empirical single-equation model restricts θ to a time-invariant constant. Evaluating whether this parameter invariance holds historically establishes the exact structural implications of the fixed-parameter closure.

3.4 Trend stabilization and the fixed-parameter closure

Under the unbalanced growth closure ($g_{Y^p} = \theta g_K$), the dynamic trajectory lacks a feasible long-run equilibrium. As established in the phase dynamics, a system experiencing over-accumulation ($\theta < 1$) forces the net accumulation rate to decay to zero. To prevent this collapse, the empirical specification in [Shaiikh \(2016\)](#) introduces a deterministic level trend (b), extending the productive-capacity closure ($g_{Y^p} = \theta g_K + b$).

Substituting this extended closure into the acceleration identity transforms the system into a

quadratic ordinary differential equation:

$$\frac{d\hat{k}}{dt} = (\hat{k} + \delta) \left[(\theta - 1)\hat{k} + b \right]. \quad (8)$$

This specification shifts the equilibrium structure of the accumulation process. The stagnation limit ($\hat{k}^* = 0$) is replaced by a new interior root:

$$\hat{k}^* = \frac{b}{1 - \theta}. \quad (9)$$

Under over-accumulation ($\theta < 1$) with positive autonomous technical progress ($b > 0$), this equilibrium is locally stable. The depreciation rate (δ) anchors the negative root, while the ratio $b/(1 - \theta)$ scales autonomous progress against the structural imbalance.

Stabilizing an unbalanced accumulation path through an exogenous algebraic term imposes a rigid theoretical restriction. The trend-stabilized closure forces a deterministic parameter (b) to offset structural imbalances, rather than endogenizing the institutional mechanisms regulating capacity formation. If technological change responds dynamically to class struggle and the wage share, the stabilizing attractor is a regime-dependent outcome of distributive conflict. Treating this algebraic stabilization as a behavioral law leaves the actual stabilization mechanism undefined (Shaikh, 1974).

The empirical validity of the constructed utilization series relies strictly on establishing a valid cointegrating vector between output, capital, and this deterministic trend. Omitting distributive conflict while including the exogenous trend introduces a double misspecification (Basu, 2020). Because the trend mathematically absorbs unobserved institutional shifts, this indirect bias channel generates plausible misleading inferences regarding the existence and stability of the cointegrating relation itself. If the deterministic trend acts as an accidental proxy for regime changes, the fundamental statistical requirement of the single-equation specification is structurally compromised.

Estimating this relation over a multi-decade sample imposes a rigid temporal constraint. This temporal restriction forces both the elasticity (θ) and the trend (b) to act as time-invariant constants across shifting institutional environments. The classical long-run closure ($\mu = 1$) isolates theoretical capital-to-capacity relations (Foley, 1985) or smooth reproduction paths (Okishio, 2022; Basu, 2022b). Because real capitalist economies routinely violate these strict proportionality conditions, imposing time-invariance forces the recovered parameters to mathematically average across structurally heterogeneous historical periods.

Looking at the actual post-war historical trajectory of the US economy (1947–2011), we see that these theoretical regimes correspond to distinct accumulation phases. During the Post-War Golden Age (1947–1973), the US economy experienced rapid capital accumulation alongside strong GDP growth and stable capacity utilization, averaging around 3.5% to 4% annual growth. This period represents a relatively balanced accumulation path. However, after the crisis of the mid-1970s and through the neoliberal era (1983–2011), the growth rate of the capital stock slowed down, and output growth decoupled from capital accumulation. The rate of capital accumulation fell to around 2% annually, while capacity utilization remained volatile, indicating a structural shift in the underlying transformation elasticity θ . Imposing a time-invariant parameter over this entire 64-year span averages across these distinct historical periods, masking the structural shifts in how capital accumulation translates to productive capacity.

The parameter θ acts as a theoretical placeholder. Section 4 implements an analytical stress test

across three distinct stages to map the conditions under which this single-equation estimation fails. Stage S0 evaluates baseline recoverability using the exact historical data. Stage S1 maps the entire admissible specification space across a 500-model ARDL grid to evaluate parameter sensitivity to information-criterion penalty schedules. Stage S2 estimates the relationship using a multi-equation Vector Error Correction Model (VECM) to determine whether joint system stability requires the logged rate of exploitation ($e_t = \ln(\pi_t/(1 - \pi_t))$) to enter the long-run cointegrating space. This sequence isolates the exact boundary conditions of the single-equation approach to preserve its portable features while establishing structural validity.

4 Critical Replication Study on Shaikh’s Approach to Cointegration

The empirical test now turns to the object defined in Section 3: Shaikh’s long-run output–capital coefficient, which draws the productive-capacity path and is interpreted here as a candidate transformation elasticity only after reconstruction. The question is whether that relation itself, not only the utilization proxy derived from it, remains stable across admissible specifications and joint system estimation.

The three stages translate this object into empirical tests. S0 reconstructs the closest recoverable version of Shaikh’s single-equation procedure. S1 opens the ARDL specification grid and separates statistical admissibility from economic admissibility. S2 moves from conditional single-equation estimation to joint VECM estimation, asking whether the bivariate output–capital relation is self-sufficient or survives under the retained criteria only when logged exploitation and historical shock controls enter the trivariate state vector.

The results narrow the conditions under which Shaikh’s strategy can carry a structural interpretation. The single-equation coefficient is recoverable, but it does not replicate the original specification exactly and is non-unique within the screened ARDL space. In joint-system estimation, the bivariate relation fails the retained system-admissibility gates; a restricted trivariate system survives under the retained criteria with distributional conditioning. The evidence therefore locates the boundary conditions of Shaikh’s portability strategy rather than discarding it.

4.1 ARDL Estimation Architecture

Section 3.1 establishes that the long-run output–capital coefficient draws the productive-capacity path. This subsection operationalizes that coefficient within an autoregressive distributed lag framework, translating the accounting identity into a testable level relationship. The ARDL bounds-testing procedure (Pesaran et al., 2001) (PSS) evaluates the existence of a cointegrating relationship between output and capital regardless of their individual integration orders, eliminating pre-testing requirements for unit roots or cointegration rank.

The procedure estimates an unrestricted equilibrium correction model derived from a general ARDL(p, q) specification:

$$\Delta y_t = c_0 + c_1 t + \phi_1 y_{t-1} + \phi_2 k_{t-1} + \sum_{i=1}^{p-1} \gamma_i \Delta y_{t-i} + \sum_{j=0}^{q-1} \delta_j \Delta k_{t-j} + \sum_{h=1}^H \delta_h D_{h,t} + \varepsilon_t. \quad (10)$$

Equation (10) writes the ARDL specification in unrestricted error-correction form, where c_0

represents the intercept and $c_1 t$ the linear trend. The short-run dynamics are represented by Δy_{t-i} and Δk_{t-j} , which capture the short-run memory of the model. The lagged level terms y_{t-1} and k_{t-1} carry the long-run content of the specification. The year-dummy block represents deterministic components that are excluded from the long-run relation $\sum_{h=1}^H \delta_h D_{h,t}$. This representation therefore separates short-run adjustment from the level relation between output and capital.

The asymptotic distribution of the F -statistic under H_0 depends explicitly on how deterministic constraints are imposed. [Pesaran et al. \(2001\)](#) identify five configurations (Cases I–V) for this constraint: Case I excludes both intercept and trend. Case II restricts the intercept to the cointegrating space. Case III includes an unrestricted intercept. Case IV includes an unrestricted intercept and restricts the trend coefficient to the null. Case V includes unrestricted intercept and trend. Each case shifts the critical value bounds because the null hypothesis alters the deterministic trending behavior of the level process. The bounds test rejects the null when the computed F -statistic exceeds the upper critical bound ($I(1)$), fails to reject when it falls below the lower bound ($I(0)$), and yields inconclusive inference when it lies within the interior band. The replication treats these deterministic configurations as admissible specification toggles. Systematic variation across Cases I–V, combined with lag-order selection and information-criterion penalties, defines the admissible specification grid.

The Wald F -statistic tests the joint exclusion restriction in equation (10). Under H_0^F , the lagged level terms do not enter the conditional model, so the specification contains no evidence of a long-run output–capital relation. Rejection of H_0^F suggests that y_{t-1} and k_{t-1} jointly contribute to the error-correction representation. The statistic is evaluated against the lower and upper bounds associated with an specific PSS deterministic case.

$$H_0^F : \phi_1 = \phi_2 = 0, \quad H_a^F : (\phi_1, \phi_2) \neq (0, 0). \quad (11)$$

where ϕ_1 and ϕ_2 are the coefficients on the lagged level terms in the unrestricted error-correction representation. Under the null, the output–capital equation contains no long-run level relation. The distribution of the statistic is non-standard because the PSS procedure brackets the polar cases in which the regressors are purely $I(0)$ or purely $I(1)$. Values below the lower bound fail to reject the absence of a long-run relation; values above the upper bound support cointegration; values between the bounds remain inconclusive.

The t -bounds statistic provides a stricter diagnostic where the deterministic case makes it formally applicable. It tests

$$H_0^t : \phi_1 = 0, \quad H_a^t : \phi_1 < 0. \quad (12)$$

Against the one-sided alternative $\phi_1 < 0$, where ϕ_1 is the coefficient on the lagged dependent variable in the conditional error-correction equation. Rejection supports an error-correction interpretation: deviations from the estimated long-run relation are followed by adjustment in subsequent periods. Because the sample is short, I evaluate the bounds evidence using the finite-sample critical values following [Natsiopoulos and Tzeremes \(2022\)](#), rather than relying only asymptotic PSS tables.

[Shaikh \(2016\)](#) operationalizes the ARDL framework to recover a single long-run output–capital multiplier, treating the coefficient as an accounting artifact rather than a production-theoretic parameter ([Shaikh, 2016](#), Appendix 6.7). The baseline specification imposes four diagnostic constraints: (1) lag orders p and q follow the Akaike Information Criterion to mitigate residual serial correlation while avoiding over-parameterization; (2) deterministic treatment adopts PSS Case IV, retaining an unrestricted intercept while restricting the trend coefficient under the null;

(3) year dummies (d_{56} , d_{74} , d_{80}) absorb outlier-driven residual spikes identified through squared-error diagnostics; (4) the Breusch–Godfrey LM test screens the ECM for autocorrelation before bounds inference proceeds. Where the F -statistic falls within the inconclusive interior band, the procedure defaults to Engle–Granger two-step estimation to establish integration order. This baseline architecture stabilizes the fitted capacity path but leaves the fixed-parameter closure assumption untested. Shaikh estimates d without theorizing its structural role or evaluating coefficient stability across distributive regimes. The replication treats this recovered multiplier as the empirical candidate for $\hat{\theta}$, routing the fixed-parameter closure assumption directly into three-stage admissibility gates.

Upon confirmation of a level relationship, the long-run coefficient recovers from the short-run ECM parameters:

$$\hat{d} = \frac{\sum_{j=0}^{q-1} \hat{\delta}_j}{1 - \sum_{i=1}^{p-1} \hat{\gamma}_i}. \quad (13)$$

In Shaikh’s procedure, this long-run multiplier anchors the fitted capacity path. Thus, capacity utilization is constructed as the residual deviation from the long-run cointegration vector: $\log \hat{\mu}_t = y_t - \hat{y}_t^p$. This sequence establishes a strict dependency: utilization derives entirely from the estimated coefficient. Shaikh treats $\hat{d}$ as a fitted accounting multiplier; he does not theorize it as a structural elasticity. This replication interprets $\hat{d}$ as the estimated candidate transformation elasticity $\hat{\theta}$ only after the theoretical mapping from capital accumulation to productive-capacity formation has been stated. Given the five ARDL deterministic cases and the year dummies used by Shaikh to control for historical outliers, the replication treats researcher decisions over specification grids, deterministic treatment, and admissibility gates as sources of variation. It stress-tests this assigned interpretation across three stages: reproducibility ($S0$), admissible-specification analysis ($S1$), and system cointegration ($S2$). Subsection 4.2 details the data construction and measurement conventions required to execute these stages.

4.2 Data and Measurement

The replication uses annual US corporate sector data, 1947–2011 ($T \approx 65$), sourced from Shaikh (2016) data companions via Real Economic Analysis.² Capacity utilization is extracted as the bounded residual deviation from the cointegrated output–capital path; the fitted capacity path inherits its level, timing, and persistence from the measurement choices detailed below. Table 2 summarizes the core variables, their construction protocols, and the measurement constraints that structure the replication.

²Replication data and construction scripts are available at Real Economic Analysis: <https://www.realecon.org/data>. Shaikh (2016) specifies the common-deflator condition but does not name the price index; the replication implements this using a GPIM-consistent implicit price deflator to ensure reproducibility.

Table 2: Core replication variables: symbols, sources, and transformations

Symbol	Construction	Role	Measurement constraint
$GVAcorp_t$	NIPA Table 1.14 (L1) + imputed-interest adj. (Table 7.11)	Output flow	Classical surplus correction
$KGCcorp_t$	BEA Fixed Asset 6.1 (L9) + IRS SOI + GPIM recursion	Capital anchor	GPIM stock-flow reconstruction
p_t	pKN implicit price deflator, 2005=100	Common price basis	GPIM-consistent deflation
y_t	$\ln(VAcorp_t/p_t)$	Dependent variable	Inherits output and deflator choices
k_t	$\ln(KGCcorp_t/p_t)$	Regressor	Inherits GPIM depletion rates
D_{56}, D_{74}, D_{80}	Squared-residual diagnostics	Disturbance controls	Baseline outlier absorption

Note: Stage $S2$ introduces $\ln e_t$ (logged exploitation) to the trivariate state vector. Distributional variables enter only at the system-admissibility gate. Full series definitions, sources, and transformations are provided in Appendix 12.

Capital stock follows the Generalized Perpetual Inventory Method (GPIM), an accounting framework that reconstructs *gross* capital stock from nominal gross investment flows, period depreciation rates, and implicit deflator built from nominal gross capital formation national accounts (Shaikh, 2016). The recursion accumulates current-cost capital as:

$$K_t = IG_t + z_t^* \cdot K_{t-1}, \quad z_t^* \equiv (1 - z_t) \frac{p_t^K}{p_{t-1}^K}, \quad (14)$$

where IG_t is nominal gross investment, z_t is the period depreciation rate, and z_t^* reflates surviving capital to current replacement cost, or depletion rate of gross capital stock. Three adjustments anchor Shaikh’s series: (1) BEA 1993 finite service lives replace the BEA 2011 infinite-lives assumption; (2) initial values are anchored to 1925 via BEA 1993 benchmarks; (3) the IRS corporate book-value index rescales the historical stock over 1925–1947 to account for Great Depression-era scrapping (Shaikh, 2016, Appendix 6.7, Sec. V). The resulting series tracks the BEA official net stock with a 99.60% average ratio over 1947–2005, confirming that level shifts reflect methodological choices rather than estimation error. Full recursion inputs and depletion-rate construction are provided in Appendix B.4.

Output and capital are deflated by the pKN implicit price index (2005=100), a GPIM-consistent implicit price deflator estimated under the same accumulation rules. Current BEA releases employ quality-adjusted chain weights that adjust for quality vintage, breaking stock-flow consistency for accumulation accounting. The pKN index preserves stock-flow consistency, ensuring the output–capital relation reflects real productive dynamics rather than quality-adjustment artifacts (Shaikh, 2016, Appendix 6.6, Sec. II). This satisfies the common-deflator condition specified in Shaikh (2016, Appendix 6.6, Eq. 6.6.7); the estimation rationale and comparison to quality-adjusted deflators are detailed in Appendix B.5.

Corporate gross value added is corrected for imputed financial intermediation services by subtracting net monetary interest paid and adding back imputed net interest adjustments (NIPA Table 7.11). The algebraic derivation and NIPA account usage are detailed in Appendix B.2. Year dummies identified by Shaikh as residual spikes enter as deterministic controls D_{56}, D_{74}, D_{80} in the ARDL baseline, following squared-error diagnostics (Patterson, 2000). They are treated here as historically interpretable shock controls associated with 1956, 1974, and 1980, not as standalone causal explanations of those episodes. The sample size ($T \approx 65$) is small for time-series inference and can bias cointegration inference when relying on asymptotic critical value bounds; the replication addresses this by computing exact-sample bounds calibrated to $T = 65$ via stochastic simulation (Natsiopoulos and Tzeremes, 2022).

These measurement conventions constitute the identification strategy. For the capacity question, the relevant capital stock is the gross stock of capital in operation, not the net book-value stock used more commonly in profitability accounting. The GPIM reconstruction is appropriate here because it tracks the accumulation and depletion of operating capital through z_t^* , preserving the stock–flow relation needed to estimate productive capacity. Net capital stock may be more directly connected to balance-sheet valuation and profitability measurement, but the capacity baseline requires a measure of the physical stock available for production.³ Shaikh also argues for a common deflator, which isolates real capital–capacity dynamics from price-distortion effects.

4.3 Empirical Design and Admissibility Strategy

This section outlines the progressive multi-stage macroeconometric strategy used to evaluate the productive-capacity path. Cointegration is the core economic concept anchoring this strategy. It asserts that even if individual time series (like output and capital) are nonstationary and drift over time, a stable linear combination of them can remain stationary. Economically, this means the variables share a long-run equilibrium relation and do not drift apart indefinitely. If cointegration holds, the residual after applying the cointegrating vector must be stationary. If it is nonstationary, the relationship is spurious, meaning the estimated equation lacks economic meaning.

We define "admissibility" as a sequence of filters rather than a single verdict. Statistical admissibility means that a specification passes the relevant econometric gate, such as ARDL bounds evidence or VECM rank and stability checks. Economic admissibility is narrower: the retained coefficient must produce an interpretable capacity parameter and a plausible utilization path. System admissibility refers specifically to survival under the joint VECM rank and companion-stability gates. Object admissibility asks whether the retained relation still estimates the intended capacity–capital mapping rather than a neighboring deterministic trend or shock-control object. A specification can therefore be statistically retained while failing economic admissibility.

The replication evaluates these admissibility gates across three progressive stages: - Stage S0 reconstructs the baseline single-equation procedure reported by Shaikh (2016) to test its reproducibility. - Stage S1 maps a combinatoric grid of 500 single-equation ARDL specifications. It tests how researcher choices over lag length, historical outlier dummies, and trend structures affect the estimated elasticity θ and its cointegration bounds. - Stage S2 shifts to a joint system framework using the Johansen (1991) reduced-rank VECM approach, testing whether bivariate output–capital cointegration is robust or requires the logged rate of exploitation to enter the state vector to achieve stability.

³Shaikh (2016) also argues that gross capital stocks are preferable for profitability indices. This chapter uses the narrower claim that gross GPIM-adjusted capital is appropriate for productive-capacity measurement.

4.3.1 Stage 0 (S_0): Reproducibility Test

Stage S_0 tests the empirical reproducibility of Shaikh’s results to establish a canonical point of departure for the replication and a benchmark for reconstruction. It selects Shaikh’s published single-equation output–capital benchmark as the target, fixes the dynamic lag structure at the reported ARDL(2,4) configuration, and estimates the model using the closest recoverable corporate-sector series. This procedure systematically identifies the exact variables within the published data tables, safely bypassing undocumented documentation omissions noted in the original baseline text. Historical impulse variables and alternative Pesaran, Shin, and Smith (PSS) deterministic configurations are estimated to reconstruct the reported point estimates and establish a baseline, rather than to execute an open specification search.

The empirical procedure estimates the long-run output–capital relation, constructs productive capacity from that fitted path, and derives capacity utilization from the deviation of actual output from the estimated capacity denominator. Cointegration bounds testing serves as the primary admissibility diagnostic to isolate valid level relationships from simple autoregressive dynamic filtering. This step highlights an essential technical property of the ARDL framework developed by [Pesaran et al. \(2001\)](#): while the F -bounds test remains universally applicable, the t -bounds statistic can only be formally defined under deterministic Cases 1, 3, and 5. Models tracking Case 2 or Case 4 configurations—which impose restricted constants or restricted trends within the cointegrating vector—cannot be subjected to a t -bounds confirmation gate.

4.3.2 Stage 1 (S_1): Admissible-Specification Stress Test

The stage S_1 protocol expands the conditional single-equation setup into an expansive multi-model space, generating a systematic grid of 500 estimated Autoregressive Distributed Lag (ARDL) specifications. By treating core modeling configurations as endogenous sources of structural variation, this framework evaluates how researcher choices over lag length, information criteria, and deterministic profiles affect cointegration bounds thresholds and long-run parameter identification. The resulting grid shifts the empirical focus from a singular baseline model to an entire admissible specification space, highlighting identification fragility under standard macroeconomic single-equation settings. This multidimensional grid strategy evaluates parameter sensitivity across changing information complexity indices, short-run structures, and historical outlier treatments.

To remove arbitrary lag choices, the grid lifts standard single-equation priors and permits full combinatoric variation across lag profiles such that $p, q \in \{1, 2, 3, 4, 5\}$. Every dynamic lag combination is estimated across all five deterministic cases outlined by [Pesaran et al. \(2001\)](#), paired with a nested block of four historical outlier controls: $s_0 = \emptyset$, $s_1 = \{D_{1974}\}$, $s_2 = \{D_{1974}, D_{1980}\}$, and $s_3 = \{D_{1956}, D_{1974}, D_{1980}\}$. These dummy frameworks absorb residual shocks linked to the Eisenhower recession, the 1974 oil shock, and the 1980 Volcker monetary contraction. This exhaustive permutation produces 500 raw single-equation models subject to structural admissibility gates. Formally, the complete specification space is defined as:

$$\mathcal{G}_{S1} = \{(p, q, \chi, s)\}, \quad (15)$$

where p and q represent the ARDL lag orders, $\chi \in \{1, \dots, 5\}$ indexes the deterministic configurations, and s indexes the nested baseline outlier structures.

The primary screen implements the F -bounds test to isolate valid level relationships from simple autoregressive dynamic filtering. Models qualify as admissible if their computed F -statistic rejects

the null hypothesis of no long-run level relationship at a 90% confidence level using an asymptotic significance threshold, formally:

$$\mathcal{A}_{S1}^F = \{m \in \mathcal{G}_{S1} : p_F(m) \leq 0.10\}, \quad (16)$$

where $p_F(m)$ represents the asymptotic p-value from the joint F-bounds test for specification m . Specifications that generate F-statistics below the lower $I(0)$ bound or within the inconclusive interior band are systematically excluded from the F-admissible space $\mathcal{A}_{S1}^F$.

The secondary diagnostic screen introduces the t-bounds statistic to verify dynamic error-correction stability. Because t-bounds are mathematically defined only under PSS Cases 1, 3, and 5, specifications under restricted constant or restricted trend branches (Cases 2 and 4) bypass this layer without penalization:

$$\mathcal{T}_{S1} = \{m \in \mathcal{A}_{S1}^F : p_t(m) \leq 0.10 \text{ if defined}\}, \quad (17)$$

where $p_t(m)$ is the asymptotic p-value from the t-bounds diagnostic. The resulting $\mathcal{T}$ -admissible space $\mathcal{T}_{S1}$ isolates specifications with robust evidence of a long-run relationship by overlapping the level-relation screen with an explicit test on the error-correction coefficient, rather than evaluating a joint test of the included variables. To identify high-confidence parameters, the Strong Cointegration Space ($\mathcal{C}_{S1}$) tightens the screening criteria to the 5% and 1% alpha thresholds:

$$\mathcal{C}_{S1} = \{m \in \mathcal{T}_{S1} : p_F(m) \leq 0.05 \text{ and } p_t(m) \leq 0.05\}, \quad (18)$$

which enforces robust empirical convergence properties across the remaining single-equation models.

Admissibility screens filter out invalid specifications but do not identify a unique parameter. Instead, the recovered parameters reflect a disciplined non-uniqueness that rejects the assumption of an invariant technical production law. Within the F-admissible pool, the first structural sorting layer constructs a Fit-Complexity Envelope (E_{S1}) to isolate optimal modeling trade-offs:

$$E_{S1} = \{m \in \mathcal{A}_{S1}^F : \nexists m' \in \mathcal{A}_{S1}^F \text{ with } L(m') \leq L(m) \text{ and } K(m') \leq K(m)\}, \quad (19)$$

where $L(m)$ is the lack of fit measured by $-2 \log L$, and $K(m)$ represents model complexity as measured by the total parameter count. This envelope isolates Pareto non-dominated specifications, mapping the geometric frontier where log-likelihood gains require expanding the parameter space.

The second sorting layer establishes information-criterion neighborhoods ($\mathcal{F}_j^{(0.20)}$) that act as "fattened" spaces around the geometric envelope. These neighborhoods relax strict optimizations to capture competitive models that sit slightly off the geometric Pareto frontier:

$$\mathcal{F}_j^{(0.20)} = \{m \in \mathcal{A}_{S1}^F : IC_j(m) \leq Q_{0.20}(IC_j \mid \mathcal{A}_{S1}^F)\}, \quad (20)$$

where $j \in \{\text{AIC, BIC, HQ, ICOMP, RICOMP}\}$, and $Q_{0.20}$ denotes the 20th percentile cutoff with ties retained. Traditional scalar penalty schedules (AIC, BIC, HQ) are augmented by Bozdogan's information complexity indices, evaluating parameter covariance structures and inverse-Fisher geometric information properties. The RICOMP criterion actively screens the parsimonious edge of the envelope by penalizing parameter interdependence. Long-run parameter estimates are highly dependent on deterministic treatments and information-criterion penalty schedules, meaning different criteria locate divergent regions of the fit-complexity space.

Every specification m yields a distinct parameter estimate $\hat{\theta}(m)$, which directly shapes the fitted capacity path. The constructed utilization series then emerges as the stationary component of a flow-stock system, downstream from a distributionally sensitive parameter space:

$$\hat{\mu}_t(m) = y_t - [\hat{a}(m) + \hat{b}(m)t + \hat{\theta}(m)k_t]. \quad (21)$$

Tracking $\hat{\mu}_t(m)$ across $\mathcal{A}_{S1}^F$, E_{S1} , and $\mathcal{F}_j^{(0.20)}$ highlights the direct economic consequences of statistical parameter drift.

Table 3 details the operational layout of the $S1$ specification search. Rather than optimizing for a single candidate model, this framework establishes the empirical variance of the transformation parameter across defensible single-equation structures. This single-equation core mapping is evaluated empirically in Section 4.5.

Table 3: Synthesis of Admissibility Specification Stress Tests

Stress-Test Component	Operational Layer	Screening/Organization Rule	Diagnostic Target
Cointegration admissibility	F/T-bounds sequence	$p_F(m) \leq \alpha$, $p_t(m) \leq \alpha$ (if defined) for $\alpha \in \{0.10, 0.05, 0.01\}$	Existence and significance of long-run level relationships
Specification invariance	Lag-order variation	Monitor parameter drift $\hat{\theta}(m)$ across combinatoric $p, q \in \{1, \dots, 5\}$	Stability and unique mapping of the long-run structural multiplier
Identification fragility	Deterministic cases & dummy structures	Track admissibility shifts across $\chi \in \{1, \dots, 5\}$ and $s \in \{s_0, \dots, s_3\}$	Sensitivity to deterministic components and structural trend assumptions
Information complexity & Pareto optimization	Fit-complexity envelope & IC neighborhoods	Pareto non-dominance in E_{S1} ; 20th percentile cutoff by information criterion	Organization of disciplined non-uniqueness under distinct information penalty schedules
Capacity benchmark sensitivity	Utilization path fan $\hat{\mu}_t(m)$	Map structural parameter variance to latent residual series	Macroeconomic consequence of single-equation coefficient drift on normal profit estimation

Note: This synthesis matrix outlines the nested single-equation admissibility screens, information-theoretic optimization layers, and diagnostic targets within the stage $S1$ multi-model grid. Source: Author's elaboration.

Section 4.3.3 extends the critique to a joint multi-equation system. Moving beyond single-equation constraints, the system-level analysis tests whether bivariate cointegration fails the system-admissibility gates in a pattern consistent with omitted distributional conditioning. This system-level non-self-sufficiency determines whether joint system survival requires the logged rate of exploitation (e_t) to enter the long-run cointegrating space.

4.3.3 Stage 2 (S2) System Co-Integration Test

Stage S2 evaluates whether the bilateral output–capital relation remains self-sufficient when shifted from a conditional single-equation setup to a joint multi-equation system. The baseline bivariate vector, $X_t = (\ln Y_t, \ln K_t)'$, fails the retained Johansen rank conditions and companion-matrix stability gates. To test this system-level non-self-sufficiency, the expanded framework introduces explicit distributional conditioning by integrating the logged rate of exploitation, $e_t = \ln(\pi_t/(1 - \pi_t))$, yielding the trivariate state vector $X_t = (\ln Y_t, \ln K_t, \ln e_t)'$. This trivariate system is estimated using the full [Johansen \(1991\)](#) Vector Error Correction Model (VECM) representation:

$$\Delta X_t = \Pi X_{t-1} + \sum_{i=1}^{p-1} \Gamma_i \Delta X_{t-i} + \Phi D_t + \varepsilon_t, \quad (22)$$

where Γ_i captures short-run dynamics, ΦD_t contains deterministic controls, and the long-run impact matrix is factored as $\Pi = \alpha\beta'$ to evaluate reduced-rank stability. Within the retained S2 grid, joint system survival is retained only under this formulation, where the rank condition $r = 1$ holds and the exploitation variable anchors the long-run cointegrating space (β).

To systematically test this stability, S2 implements a VECM combinatoric grid mirroring the single-equation sensitivity approach. Formally, the complete attempted specification space for each target rank (r) is defined as:

$$\mathcal{G}_{S2}^{(r)} = \{(p, \chi, s)\}, \quad (23)$$

where $p \in \{1, 2, 3, 4\}$ represents the VAR lag orders, $\chi \in \{d_0, d_1, d_2, d_3\}$ indexes the Johansen deterministic closure labels, and $s \in \{h_0, h_1, h_2\}$ isolates a subset of the baseline historical outlier controls. The labels $d_0, \dots, d_3$ denote deterministic VECM branches, not the long-run coefficient d in equation (4). Permuting across these state-vectors yields an attempted grid of 48 specifications per rank. However, because specifications under the restricted constant branch (d_1) systematically fail numerical convergence across all permutations, the successfully estimated baseline is analytically restricted to exactly 36 models per system family.

The Stage S2 admissibility protocol evaluates these surviving 36 specifications through a rigorous triple gate requiring numerical convergence, Johansen rank confirmation, and companion-matrix dynamic stability. Within the trivariate framework, the reduced-rank restriction $r = 1$ tests whether a single cointegrating relation survives distributional integration, while $r = 2$ evaluates the potential for a dual-relation system. To organize the surviving specifications, the subset Ω_{20} ranks the admissible models based strictly on minimized log-likelihood loss. This likelihood-based organization layer establishes an interpretive hierarchy for the retained systems but does not independently define the primary boundaries of system admissibility.

4.3.4 Cross-Stage Synthesis

The multi-stage critical replication executes an intentionally nested screening architecture to map the boundary conditions of the productive-capacity path. Stage S0 establishes the approximate empirical recoverability of the baseline single-equation framework. Stage S1 expands the estimation framework into a comprehensive specification grid, utilizing structural bounds diagnostics and information-criterion envelopes to map parameter non-uniqueness. Finally, Stage S2 imposes multi-equation Vector Error Correction Model (VECM) stability gates to verify whether the underlying structural relation survives when estimated jointly. Table 4 formalizes this progressive methodological sequence.

Table 4: Stage Design and Admissibility Strategy

Stage	Design Object	Specification Space	Admissibility and Organization	Interpretive Implication
S0	Single-point reconstruction of baseline specification	ARDL(2,4), historical impulse dummies, alternative deterministic closures	F-bounds primary screen; t-bounds diagnostic where defined; utilization path proximity	Tests whether a deterministic capacity path can be closely reconstructed
S1	ARDL specification grid $\mathcal{G}_{S1}$	Lag orders (p, q) , five PSS cases, four nested dummy structures	F-admissible set $\mathcal{A}_{S1}^F$; F+t confirmed candidates $\mathcal{C}_{S1}$; corrected envelope E_{S1} ; IC-specific neighborhoods $\mathcal{F}_j^{(0.20)}$	Tests whether the output-capital coefficient is structurally unique or varies with IC penalty logic
S2 Bivariate	Joint $(\ln Y_t, \ln K_t)'$ VECM	Lag order, deterministic branch, dummy structure, $r = 1$	Triple gate $\mathcal{A}_{S2}$: convergence, Johansen rank, companion stability	Tests whether the bilateral output-capital relation is self-sufficient at the system level
S2 Trivariate	Joint $(\ln Y_t, \ln K_t, \ln e_t)'$ VECM	Same as bivariate, with logged rate of exploitation included; $r = 1$ or $r = 2$	Same triple gate; Ω_{20} organizes admissible survivors by log-likelihood loss	Tests whether the structural relation survives only within the retained S2 grid when distribution enters explicitly

Note: This table outlines the progressive multi-stage replication architecture. The sequence advances from baseline single-equation reconstruction (S0) through parameter sensitivity grids (S1) to joint system-level survival testing (S2). Source: Author’s elaboration.

4.4 Stage 0 (S0): Single-Equation Reconstruction

Stage S0 provides an approximate reconstruction of the empirical baseline by comparing Shaikh’s published estimate of $\hat{\theta} = 0.66$ with the closest recoverable estimate, $\hat{\theta} = 0.72$. This initial stage recovers the empirical fitted capacity path from the nearest documented single-equation output-capital specification, focusing on point-estimate recoverability. Shaikh (2016) estimates d , the long-run output-capital coefficient, which this article interprets as $\hat{\theta}$ due to its theoretical role as the candidate transformation elasticity linking capital accumulation to productive-capacity formation, as outlined in Section 3.

The S0 exercise is a controlled reconstruction test that explicitly defines the baseline choices governing data construction, deterministic treatment, and long-run closure, given the incomplete reporting trail of result-level estimates and variable provenance in Shaikh (2016). The result establishes the closest recoverable benchmark under these documented historical impulse controls and lag structures. Notably, the borderline bounds significance of $p = 0.099$ signals the potential fragility of the single-equation specification.

The published reference value is $\hat{\theta} = 0.66$, whereas the closest recoverable reconstruction yields $\hat{\theta} = 0.72$ from an ARDL(2,4) specification, which also emerges as the BIC winner in the S0 rerun. An alternative ARDL(4,3) specification, selected by the AIC, yields $\hat{\theta} = 0.75$. The precise parameters for these two estimated S0 long-run multipliers are reported in Table 5. The discrepancy between 0.66 and 0.72 is the primary S0 result: the baseline is approximately recoverable, but not exactly reproducible under documented choices. To recover the benchmark, an active search through Shaikh’s raw spreadsheet archives was necessary to identify the undocumented deflator choice among

competing price deflator indices, such as the BEA historical-cost series vs. Census book values.

Table 5: Stage S0 Long-Run Output–Capital Multipliers

Reference/Model	Order	Role	$\hat{\theta}$	SE	t	p
Shaikh (2016) Benchmark	ARDL(2,4)	Published reference	0.66	–	–	–
Faithful Reconstruction	ARDL(2,4)	Reconstructed benchmark	0.72	0.06	11.82	< 0.001
AIC Comparator	ARDL(4,3)	Lag-length comparator	0.75	0.04	18.43	< 0.001

Note: This table reports Stage S0 long-run output–capital multipliers. Shaikh’s published estimate operates as the baseline reference value; standard errors, t -statistics, and p -values are reported exclusively for the generated replication estimates. Source: Author’s elaboration.

The bounds testing procedure in the ARDL framework evaluates the existence of a long-run relationship without prior assumptions regarding the integration order of the variables. The test brackets the polar cases where the regressors are purely $I(0)$ or purely $I(1)$ to evaluate whether a meaningful level relationship exists. Under the null hypothesis of no cointegration, the joint exclusion of lagged level variables is tested.

The computed test statistics are compared against critical values generated by stochastically simulated finite-sample distributions. In our faithful reconstruction, the F -bounds statistic yields a value of 3.349, which barely clears the $I(1)$ critical value threshold of 3.333 at the 10% significance level ($p = 0.099$). This borderline significance indicates that the bivariate single-equation relationship between output and capital is highly sensitive to minor specification parameters, such as lag selection or outlier treatment.

A small change in the estimated elasticity θ from 0.72 to 0.75 alters the constructed utilization series by as much as 5 percentage points. This sensitivity arises because the estimated parameter propagates directly through the capacity definition, where any deviation in $\hat{\theta}$ shifts the capacity ceiling and changes the resulting utilization levels.

The t -bounds statistic serves as a secondary diagnostic, enforcing a stricter error-correction stability condition. While useful for tightening the admissibility criteria, it acts as a complement to the F -bounds test. The borderline significance of the bounds tests ($p = 0.099$) underscores the sensitivity of the cointegrating space to minor changes in lag structures and deterministic terms (Pesaran et al., 2001).

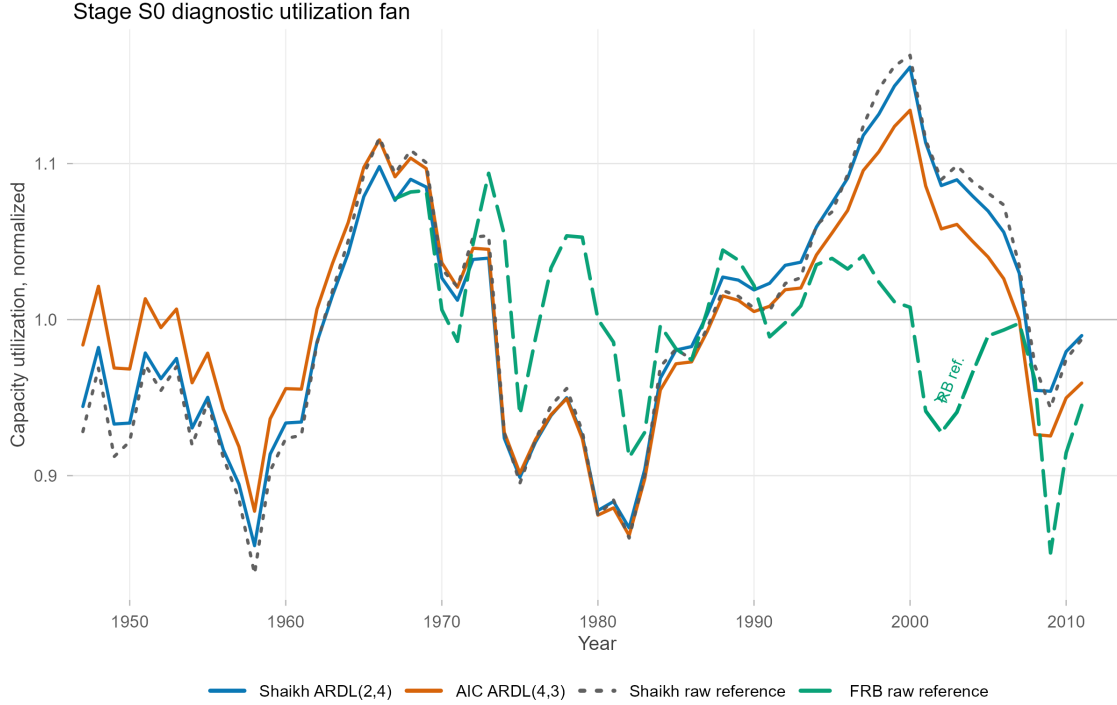
Table 6: Stage S0 ARDL Bounds Evidence at $\alpha = 0.10$ (90% Confidence Intervals) with Small-Sample Critical Values

Model	Test	PSS Case	Statistic	I(0) Bound	I(1) Bound	Exact-Sample p	Verdict
ARDL(2,4)	F -bounds	Case 1	3.349	2.466	3.333	0.099	Reject H_0
ARDL(2,4)	t -bounds	Case 1	-2.507	-1.602	-2.259	0.062	Reject H_0
ARDL(4,3)	F -bounds	Case 1	5.250	2.486	3.338	0.023	Reject H_0
ARDL(4,3)	t -bounds	Case 1	-3.125	-1.621	-2.270	0.015	Reject H_0

Note: The null hypothesis assumes no long-run level relation. The F -bounds statistic supplies the primary admissibility diagnostic. The t -bounds statistic is reported as a stricter secondary diagnostic where the deterministic structure permits. Case 1 denotes the no-constant/no-trend PSS deterministic configuration. Rejection is assessed at $\alpha = 0.10$. Source: Author’s elaboration.

Figure 2 plots the estimated capacity utilization paths generated under alternative single-equation specifications, alongside Shaikh’s published series and the Federal Reserve Board (FRB) manufacturing index. The spread between the reconstructed ARDL(2,4) benchmark ($\hat{\theta} = 0.72$) and the AIC-selected ARDL(4,3) comparator ($\hat{\theta} = 0.75$) shows that utilization estimates are highly sensitive to lag-length selection. A minor adjustment of 0.03 in the estimated transformation elasticity yields up to a 5 percentage point level shift in constructed utilization, indicating that single-equation capacity paths are highly sensitive to lag and specifications.

Figure 2: S0 Diagnostic Utilization Fan



Note: The fan maps the variance of constructed utilization paths given competing ARDL specifications. Source: Author’s elaboration.

Stage S0 confirms that a Shaikh-like capacity benchmark is approximately recoverable. However, recovering a single point estimate does not guarantee that the underlying structural parameter is unique or robust. To expose potential identification fragility, Stage S1 expands the estimation framework. This next stage tests whether the recovered benchmark remains stable across a wide grid of statistically retained single-equation alternatives.

4.5 Stage 1 (S1) Admissible-Specification Stress Test Results

Stage S1 tests the model’s stability by running a 500-regression grid search, systematically varying lag order, deterministic case, dummy structure, and information criteria to evaluate whether the baseline benchmark remains stable. In the notation of Section 4.3, we move from a single ARDL point estimate to the fully screened grid space $\mathcal{G}_{S1}$.

The resulting grid does not collapse around a single parameter, instead containing 500 successful combinatorial specifications. Applying the initial 10 percent bounds threshold, 102 specifications

pass the primary F -bounds gate and enter the statistically retained F -admissible set $\mathcal{A}_{S1}^F$. As the asymptotic significance size tightens, the retained set predictably shrinks: 62 specifications remain F -admissible at the 5 percent threshold, and just 13 at the 1 percent limit. Unpacking this set reveals that cointegration is concentrated in specific cases and dummy configurations, with no-trend specifications dominating 90.2% of the statistically retained space (Cases 1, 2, and 3) and trend-containing models representing only 9.8%. Level stability is also highly conditional on dummy variables: 92.2% of the F -admissible models require at least one structural break dummy variable (s_1, s_2, s_3) to pass the bounds test, indicating that step-shift dummies act as historically interpretable shock controls associated with 1956, 1974, and 1980.

While the t -bounds diagnostic enforces a stricter requirement for dynamic error-correction stability, the grid search demonstrates that bounds-test admissibility remains highly sensitive to lag and deterministic choices. A model-selection routine that depends entirely on the researcher’s chosen information criterion reflects the researcher’s preference for parameter penalties rather than indicating a unique cointegrating relationship.

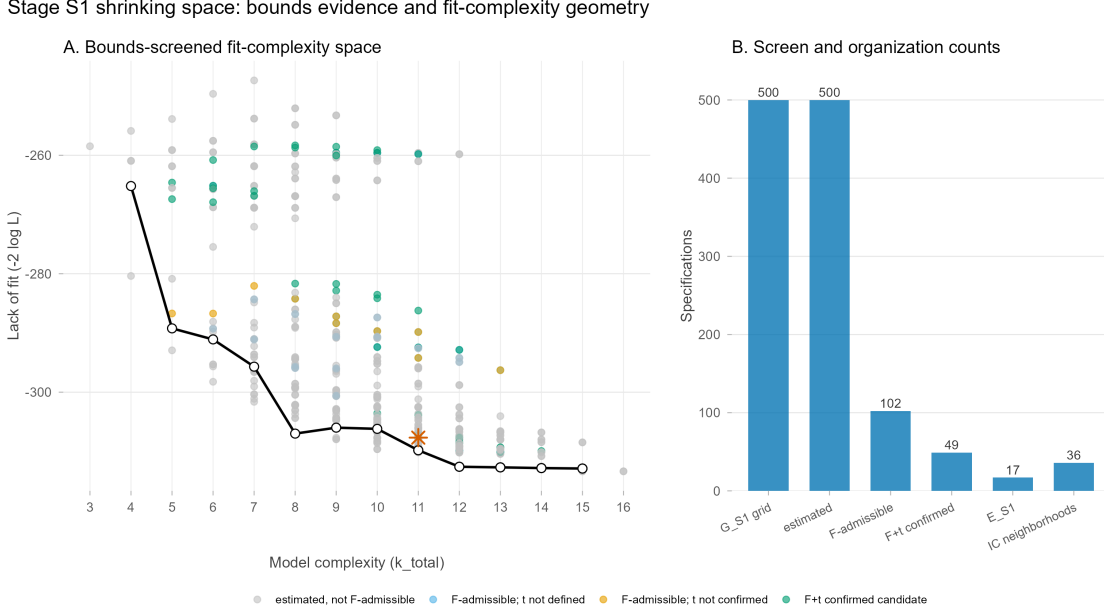
Table 7: S1 Shrinking Specification Space Counts

Layer	Count
Full grid	500
Estimated successfully	500
F-admissible at 10%	102
F+t confirmed at 10%	49
F-admissible at 5%	62
F+t confirmed at 5%	28
F-admissible at 1%	13
F+t confirmed at 1%	3
E_{S1}	17
IC-neighborhood union	36

Note: This table tracks the survival rate of the 500 estimated single-equation ARDL models through the nested cointegration admissibility gates. Source: Author’s elaboration.

Figure 3 reports this screening hierarchy visually. Panel A maps the 500 estimated specifications across the fit–complexity space, color-coded strictly by bounds-admissibility status. Panel B translates this geometry into exact specification counts surviving each sequential screening layer. This sequential ordering reinforces the fundamental methodological rule that model comparison begins strictly after the bounds screen, not before it.

Figure 3: S1 Shrinking Specification Space



Source: Author's elaboration.

Within the F -admissible set, the corrected global non-dominated fit-complexity envelope (E_{S1}) isolates 17 optimal specifications. The horizontal dimension of this geometric space measures model complexity via the total number of estimated parameters, while the vertical dimension measures the lack of fit, reported formally as $-2 \log L$. Moving along the E_{S1} frontier therefore trades statistical fit against structural parsimony. The resulting envelope serves strictly as a geometric summary of Pareto-optimal ARDL alternatives, not as a singular new estimator.

This strong statistical dependence on dummy variables aligns with the historical trajectory of the postwar US economy. The shock years correspond to major systemic disruptions: the 1956 Eisenhower recession, the 1974 first OPEC oil crisis and stagflation, and the 1980 Volcker monetary shock. In a pure bivariate level relation without outlier controls (s_0), these historical episodes appear as persistent, non-mean-reverting shocks in the residuals. In those specifications, the error term behaves as a random walk, violating stationarity and causing the bounds test to fail. Retained bounds evidence is obtained when step-shift dummy variables (s_1, s_2, s_3) absorb level shifts associated with these historical shocks and allow the error term to return to a zero-mean stationary path. The negative coefficients on these dummies are interpreted here as level adjustments in capital productivity (Y/K) associated with these crises. The dummies are therefore historically interpretable shock controls, not independent evidence of the full causal mechanism behind each episode.

The IC-specific neighborhoods operate as post-admissibility, criterion-conditioned organization layers. For each criterion $j \in \{\text{AIC}, \text{BIC}, \text{HQ}, \text{ICOMP}, \text{RICOMP}\}$, the neighborhood $\mathcal{F}_j^{(0.20)}$ isolates the bottom 20 percent of F -admissible specifications, capturing 36 unique models across the union. Table 8 reports the precise parameter ranges across these respective criterion borders, demonstrating that these neighborhoods organize the retained specifications rather than establish cointegration natively.

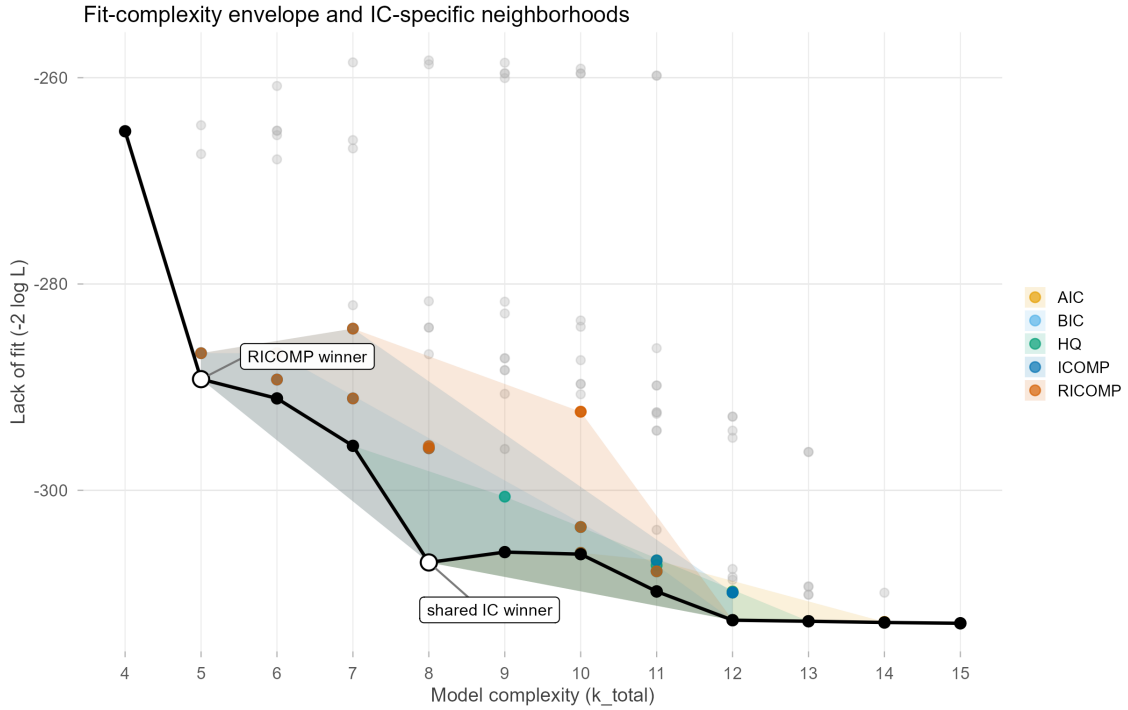
Table 8: S1 IC-Specific Neighborhood Summaries

Criterion	Specs	Min $\hat{\theta}$	Mean $\hat{\theta}$	Median $\hat{\theta}$	Max $\hat{\theta}$
AIC	22	0.84	0.88	0.86	0.93
BIC	21	0.65	0.91	0.86	2.20
HQ	21	0.75	0.94	0.86	2.20
ICOMP	22	0.65	0.84	0.86	0.95
RICOMP	22	0.65	0.83	0.85	0.95

Note: The table reports the summary statistics for the long-run parameter $\hat{\theta}$ within the bottom 20th percentile of each information criterion. Estimates are rounded to two decimal places for consistency. Source: Author's elaboration.

Figure 4 supplies the primary fit-complexity and IC-neighborhood evidence. The solid black line denotes the Pareto non-dominated fit-complexity envelope (E_{S1}). Surrounding this frontier, the shaded polygons define the 20th percentile neighborhoods for each respective information criterion. This visualization maps the spatial divergence between the highly parsimonious RICOMP winner and the shared scalar-penalty winner.

Figure 4: S1 Fit-Complexity Envelope and IC-Specific Neighborhoods



Source: Author's elaboration.

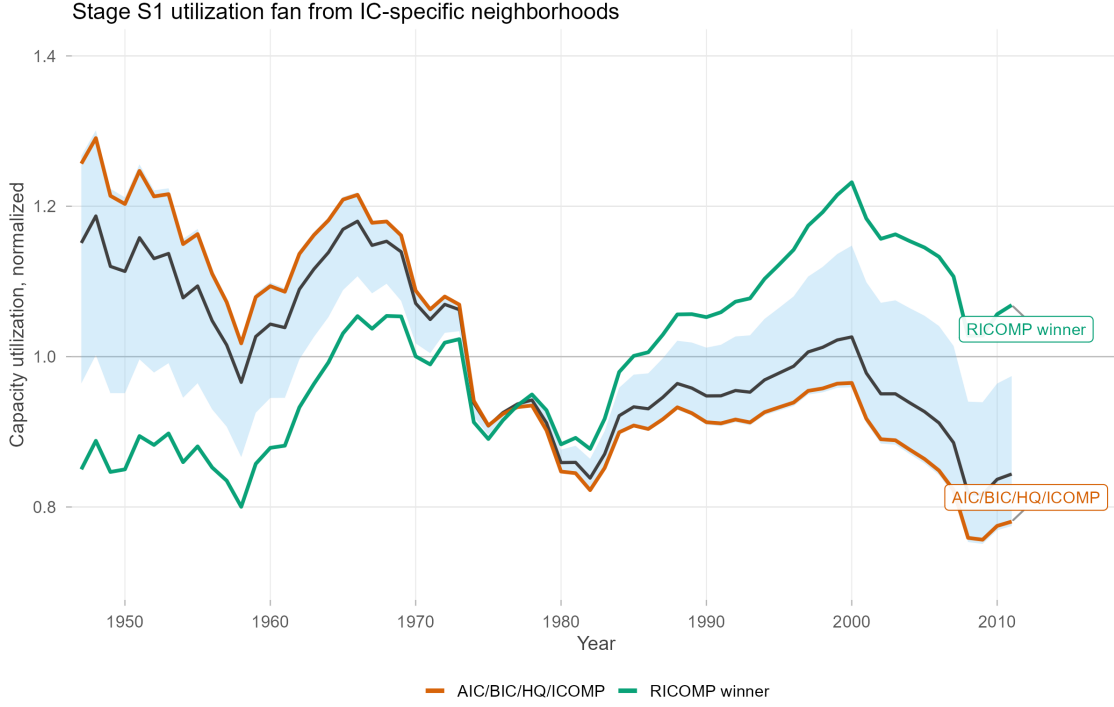
The shared AIC/BIC/HQ/ICOMP winner is an ARDL(3,3), Case 1 specification with the 1974 historical dummy (s_1), yielding $\hat{\theta} = 0.92$. Conversely, RICOMP selects a highly parsimonious ARDL(1,2), Case 2 specification with no impulse dummies (s_0), yielding $\hat{\theta} = 0.65$. This criteria

divergence does not establish a structural ranking of the underlying estimators. Unlike standard scalar penalties, the ICOMP and RICOMP criteria evaluate complexity through the covariance structure of estimated parameters and residual dependence (Bozdogan, 1990, 2000). The RICOMP neighborhood therefore marks a covariance-complexity-sensitive region of the admissible space rather than providing a definitive model-selection verdict (Güney et al., 2021). This parameter variation also matters for economic interpretation. While no-trend specifications (Cases 1, 2, 3) yield stable long-run elasticities bounded below unity ($\hat{\theta} \in [0.65, 0.95]$) that support the overaccumulation hypothesis, the trend-containing configurations (Cases 4, 5) yield economically inadmissible capacity-benchmark magnitudes ($\hat{\theta}$ averaging 1.36 and reaching up to 2.20). An elasticity $\theta > 1.0$ implies that productive capacity expands exponentially relative to capital accumulation in the long run, violating the classical overaccumulation baseline and indicating that statistical bounds-passing does not guarantee economic admissibility.

Figure 5 serves a narrower evidentiary role by reporting the diagnostic constructed-utilization paths generated strictly from the retained IC-neighborhood specifications. The shaded region maps the variance of constructed capacity utilization, while the divergent paths of the distinct criteria winners highlight the macroeconomic sensitivity to single-equation penalty schedules. These paths provide supporting evidence for parameter non-uniqueness, demonstrating how the implied utilization series mechanically shifts when the capacity denominator relies on different retained ARDL alternatives.

To evaluate this single-equation parameter sensitivity, consider a key pedagogical counterfactual: the restricted balanced-growth model where the transformation elasticity is constrained to unity ($\theta = 1.0$). If a researcher forces this standard Harrodian assumption onto the US dataset, the capacity path is forced to expand exactly in step with the capital stock (imposing a constant capital productivity baseline). When we construct capacity utilization under this constraint, the resulting series drifts to economically implausible levels, either falling continuously toward zero in the long run or exceeding 120% during periods of rapid expansion. This occurs because the economically admissible estimates imply an unbalanced US output–capital relation ($\theta < 1.0$), meaning capacity grows more slowly than the capital stock. Imposing the $\theta = 1$ counterfactual falls outside the retained economic-admissibility interpretation, showing why estimating the unrestricted parameter is not just a technical choice, but a requirement for economic viability.

Figure 5: S1 Diagnostic Utilization Fan from IC-Specific Neighborhoods



Source: Author's elaboration.

Stage S1 demonstrates that the statistically retained single-equation space is populated, yet it does not support the assumption of a unique structural benchmark. Because the retained alternatives fail to collapse into a single transformation elasticity ($\hat{\theta}$), the resulting single-equation utilization path remains structurally fragile. Section 4.3.3 addresses this fragility by testing whether the output–capital relation survives multi-equation system estimation, and whether survival under the retained criteria depends on distributional variables entering the state vector.

4.6 Stage 2 (S_2) System-Level VECM Replication

Stage S2 extends the empirical analysis to a system-level vector error correction model (VECM) to evaluate the joint dynamics of output and capital. The bivariate state vector, $X_t = (\ln Y_t, \ln K_t)'$, is examined first. However, the bivariate system does not pass the retained system-admissibility gates across all estimated bivariate specifications. The VECM residuals remain nonstationary, indicating that the long-run relation between output and capital is not system-admissible when estimated in isolation.

To address this issue, the trivariate state vector $X_t = (\ln Y_t, \ln K_t, \ln e_t)'$ is introduced, where $\ln e_t$ represents the logged rate of exploitation, constructed as the aggregate corporate profit-share-to-wage-share ratio, $e_t = \pi_t / (1 - \pi_t)$. The inclusion of $\ln e_t$ allows the test to place distributional variables within the cointegrating space. The system-level estimation is governed by a triple confirmation gate, consisting of numerical convergence, Johansen reduced-rank verification, and companion-matrix dynamic stability, to isolate retained long-run systems from unstable vector autoregressive processes.

The results are presented in Table 9, which details the structural outcomes across the complete VECM specification space ($\mathcal{G}_{S2}$). The bivariate $r = 1$ configuration yields zero system-admissible models, with VECM residuals that fail unit-root tests and remain nonstationary. In contrast, the trivariate $r = 1$ framework yields six statistically retained system specifications when conditioned on the logged rate of exploitation, $\ln e_t$. The trivariate rank-two permutations confirm this restriction, producing zero stable dual-vector systems. This indicates that cointegration is retained only within the rank-one trivariate system in the estimated S2 grid, alongside step-shift dummies. The six retained systems must then pass a second economic-admissibility screen before they can be treated as valid capacity benchmarks.

Table 9: S2 System Admissibility Outcomes

System	Rank	Estimated	Admissible	Interpretation
Bivariate ($\ln Y, \ln K$)	$r = 1$	36	0	Output–capital relation is not self-sufficient
Trivariate ($\ln Y, \ln K, \ln e$)	$r = 1$	36	6	Restricted survival with logged exploitation
Trivariate ($\ln Y, \ln K, \ln e$)	$r = 2$	36	0	No admissible two-vector system

Source: Author’s elaboration.

The h_2 dummy vector plays a central role in the trivariate VECM models, as it accounts for the step shifts in 1956, 1974, and 1980, and it appears in each retained system-admissible specification. The uniform survival pattern of models that include this vector indicates that the single-equation outlier controls capture level shifts associated with system-level stability. Cointegration is retained only when these historically interpretable shock controls prevent persistent level shifts in the residual mean from appearing as unit roots in Johansen trace and Dickey-Fuller tests; the dummies do not by themselves identify the causal mechanisms behind each historical episode. Under the d_3 trend branch, which includes a linear time trend in the VECM cointegrating space, the models yield economically inadmissible parameter estimates, including explosive ($\hat{\theta} = 11.31$) or negative ($\hat{\theta} = -0.81$) elasticities, because the time trend absorbs the level drift of output, forcing the capital-output coefficient to capture short-run noise and structural breaks. These d_3 configurations are therefore economically rejected as capacity benchmarks. The $p2.d0.h2.r1$ specification is also statistically retained but economically rejected as a capacity benchmark because its $\hat{\theta} = 1.19$ lies above unity, reversing the chapter’s economically interpretable overaccumulation closure. The economically admissible S2 benchmarks are therefore restricted to the positive below-unity estimates $\hat{\theta} = 0.73, 0.91, 0.97$. The negative coefficients on the shock controls are interpreted as level shifts associated with 1956, 1974, and 1980, not as proof of a complete historical mechanism. The logged rate of exploitation e_t is also associated with greater system-level stability. By incorporating the distribution variable into the state vector, the retained S2 design is consistent with a Sraffa-style choice-of-technique channel in which institutional distribution conditions the stable long-run output–capital relationship (Kurz, 1986).

The no-dummy counterfactual specification provides a key test of the importance of historical controls. Omitting the h_2 dummy vector, which controls for the 1956, 1974, and 1980 shocks, results in a VECM that fails to reject the null hypothesis of no cointegration, according to the Johansen trace and maximum eigenvalue tests. Moreover, the resulting VECM residuals are highly

nonstationary, failing standard unit-root tests. This indicates that the estimated relationship does not pass the retained system-admissibility gates without an error-correction mechanism to pull output, capital, and exploitation back together. The counterfactual indicates that the system-level output–capital relationship is not retained when specific historical and institutional shock vectors are excluded. Under the retained S2 design, the cointegrating space is therefore conditional on these historical dummy variables rather than a permanent, institution-free law.

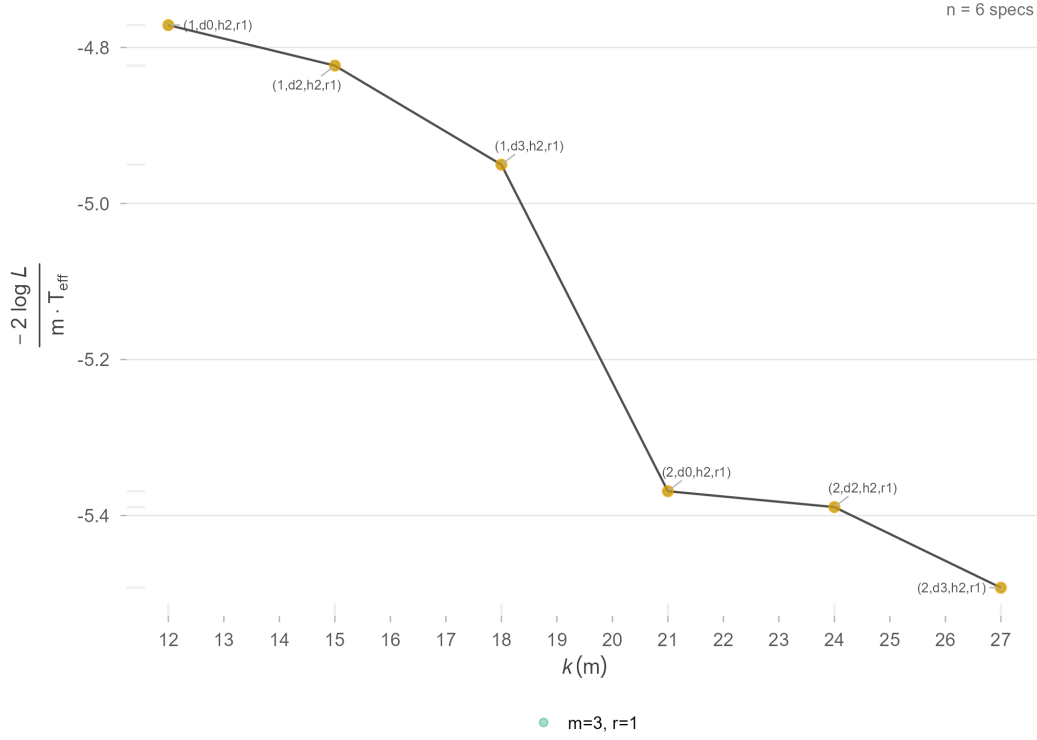
Table 10: S2 Retained Trivariate Specifications

Lag Order (p)	Deterministic Branch Label	Historical Dummy	Rank (r)	Cointegrating Coefficient ($\hat{\theta}$)
1	d_0	h_2	1	0.91
1	d_2	h_2	1	0.97
1	d_3	h_2	1	-0.81
2	d_0	h_2	1	1.19
2	d_2	h_2	1	0.73
2	d_3	h_2	1	11.31

Note: Retained means the specification passes the S2 statistical/system gates. Economic admissibility is imposed separately; the d_3 estimates and the above-unity $p2_d0_h2_r1$ estimate are not treated as valid capacity benchmarks. Source: Author’s elaboration.

Figure 6 plots the specification frontier mapping system-level fit against parameter complexity for the trivariate VECM specifications. The results show that statistically retained parameters are constrained to a narrow region, while the economic-admissibility screen narrows the usable capacity benchmarks further. Bivariate systems do not pass the retained system-admissibility gates, and expanding the state vector beyond the trivariate specifications does not improve system-level identification, indicating that the trivariate system is the minimum retained specification in this design.

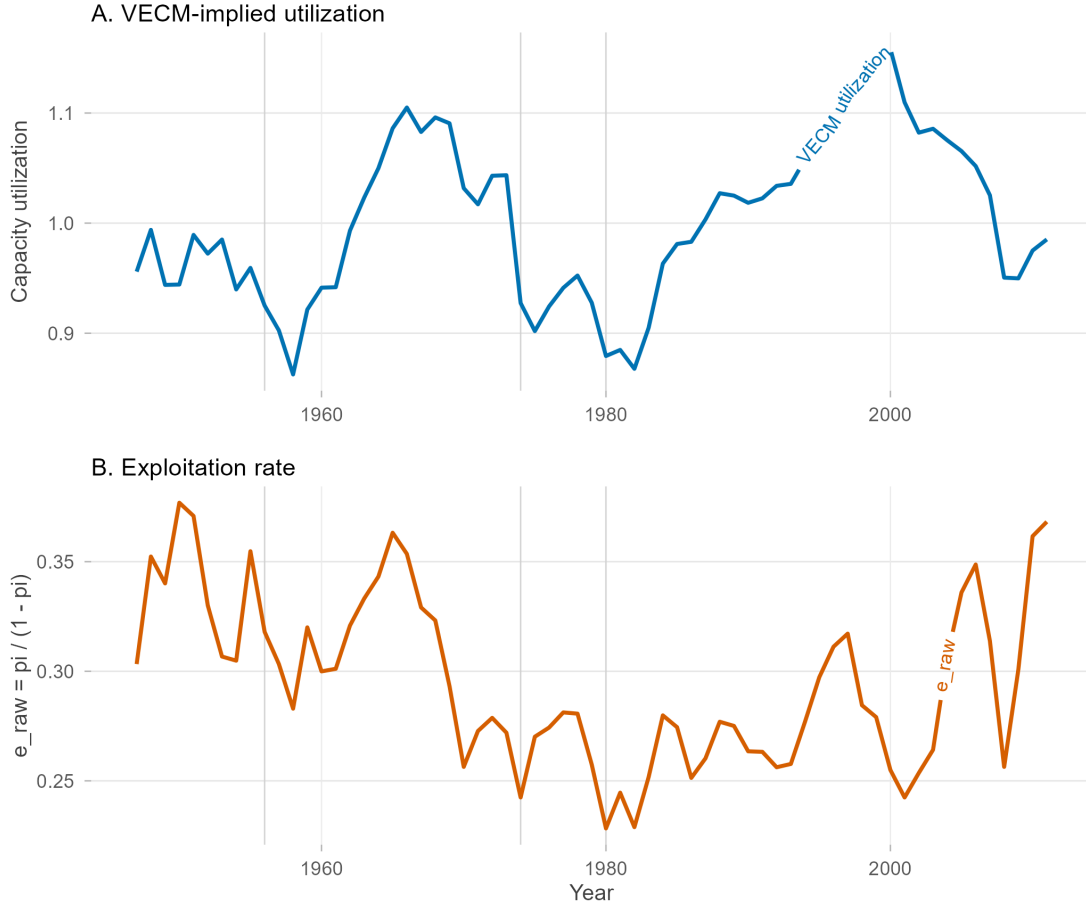
Figure 6: S2 Pooled System Frontier



Source: Author's elaboration.

Figure 7 shows how the system's final state changes over time. The left panel plots the path of capacity utilization implied by the Vector Error Correction Model, while the right panel charts the actual rate of exploitation over time. In the US, the rate of exploitation reached its highest point in the mid-1960s, during a period of strong economic growth after World War II. It then dropped during the economic stagnation and high inflation of the 1970s, but rose again from 1983 to 2011 as companies increased productivity faster than wages. This long-term pattern is consistent with the retained S2 interpretation that capacity-utilization measurement is distributionally conditioned by the functional distribution of income between wages and profits.

Figure 7: S2 Focal Utilization and Exploitation Diptych



Source: Author's elaboration.

Heinz Kurz's (1986) classical choice-of-technique switching argument offers a mechanism through which changes in distribution, specifically the rate of exploitation e_t , alter the relative cost-effectiveness of different production techniques, prompting cost-minimizing producers to switch their technical methods and adjust their capital utilization, thereby endogenizing capacity. As producers respond to shifts in income distribution by changing their production techniques, the observed output-capital relation (Y/K) may become sensitive to distributional conditions rather than operating as a fixed technical law (Kurz, 1986). The S2 evidence is consistent with this mechanism: the bivariate system does not retain a stable cointegrating vector under the retained gates, while the trivariate system is statistically retained only when logged exploitation and historical shock controls enter the state vector. Within this design, capacity identification is therefore a distributionally conditioned capacity parameter rather than a pure technical engineering relation.

4.7 Cross-Stage Econometric Synthesis

The empirical findings across the three estimation layers support a bounded critical-replication result. Stage S0 shows that the baseline capacity path is approximately recoverable, but not

exactly reproducible under documented choices. Stage S1 scales this reconstruction into a non-empty statistically admissible ARDL space of disciplined non-uniqueness, where the long-run parameter varies along a Pareto fit–complexity frontier and some retained specifications fail economic admissibility. Stage S2 imposes joint multi-equation validation, showing that the bivariate output–capital relation is not self-sufficient and that restricted trivariate survival occurs under the retained criteria when logged exploitation and historical shock controls enter the long-run system.

Table 11: Cross-Stage Synthesis of the Cointegration Architecture

Stage	Outcome	Interpretive Bound
S0	Approximate single-equation reproducibility	Shaikh’s method is reproducible, but not exactly
S1	Non-empty statistically admissible ARDL grid; non-unique retained benchmark	Model selection is driven by information-criterion choice
S2 Bivariate	36 estimated, 0 system-admissible	Output–capital system is not self-sufficient
S2 Trivariate Rank One	36 estimated, 6 statistically retained; 3 economically admissible	Restricted survival with logged exploitation and h_2 controls
S2 Trivariate Rank Two	36 estimated, 0 system-admissible	No retained dual-vector system

Source: Author’s elaboration.

The magnitude of the estimated transformation elasticity ($\hat{\theta}$) dictates both the historical baseline of the capacity utilization path and its dynamic mean-reverting properties. Under Shaikh’s methodology, utilization is constructed strictly as the residual distance between realized output and fitted capacity ($\ln \hat{\mu}_t = y_t - \hat{y}_t^p$). The estimated $\hat{\theta}$ therefore determines exactly how much of the smooth, long-run capital trend is subtracted from volatile current output. Lower parameter estimates force the residual to absorb structural drift, generating a utilization path characterized by long historical waves and dampened mean reversion. Conversely, higher parameter estimates aggressively strip the capital trend from the residual, resulting in a highly stationary trajectory that mirrors short-run business cycle volatility.

Multi-equation vector estimation narrows this single-equation ambiguity by identifying the historical and institutional conditions under which structural stability is retained. The six statistically retained trivariate VECM specifications include the h_2 historical dummy vector, which accounts for the structural breaks of 1956, 1974, and 1980. This uniform survival pattern indicates that the outlier controls implemented in Shaikh’s baseline operate as system-level conditions for the retained cointegrated space, rather than arbitrary statistical adjustments. Cointegration is retained only when these historically interpretable shock vectors are included in the long-run correlations between accumulation and production.

The non-self-sufficiency of the standalone output–capital relation is consistent with the dual macroeconomic role of capital accumulation. Because investment demand expands current output while the capital stock simultaneously builds productive capacity, the scalar parameter $\hat{\theta}$ is forced to mediate conflicting short-run and long-run forces. In a single-equation framework, this accounting

entanglement is consistent with omitted distributional conditioning and parameter instability. The Stage S2 trivariate system indicates that the empirical measurement of capacity utilization is not fully captured as a purely technical engineering problem. Productive capacity is therefore a political-economy object, trackable in this design only when the estimation framework explicitly integrates the distributional conflict captured by the logged rate of exploitation.

5 Conclusion

This chapter has reexamined Anwar Shaikh’s capacity utilization index, reconstructing the empirical and theoretical conditions required to identify the productive ceiling in the post-war U.S. economy. The replication shows that economically admissible retained specifications generally estimate the structural parameter governing the long-run output–capital relationship below unity ($\hat{\theta} < 1.0$). Some statistically retained specifications generate values at or above unity, including the S2 estimate $\hat{\theta} = 1.19$, while others produce implausible negative or explosive magnitudes. These are treated as failures of economic admissibility rather than valid capacity benchmarks. The evidence rejects the standard Harroddian balanced-growth assumption ($\theta = 1.0$) only under the retained economic-admissibility criteria, indicating that the U.S. economy can be interpreted as operating under an unbalanced growth closure where capital accumulation outpaces capacity formation. In this regime, the system exhibits a structural tendency toward overaccumulation and stagnation, whose concrete historical stabilizers require direct treatment beyond the present chapter’s empirical design.

The empirical results carry significant theoretical consequences for classical-Marxian profitability crisis models. In this tradition, the general rate of profit is decomposed as the product of the profit share, capacity utilization, and the capacity-to-capital ratio. Under the economically admissible below-unity estimates, the capacity-to-capital ratio exhibits a secular downward trend as capital accumulates. This falling capital productivity acts as a persistent downward drag on the rate of profit, reinforcing the Marxian tendency of the rate of profit to fall. The chapter does not estimate the demand components that may bound this tendency. A fuller account would need to examine whether military spending, household credit, or other autonomous demand components stabilize the capacity path across historical regimes. Our system-level analysis shows that omitting the distribution variable and historical shock vectors from the estimation yields spurious relationships, indicating that macroeconomic capacity utilization is not a neutral technical index but a variable shaped by class conflict and institutional interventions.

A key limitation of the present analysis is the treatment of the transformation elasticity θ as a time-invariant parameter. While a fixed-parameter closure is necessary for empirical estimation over the post-war period, the capacity-creating effect of capital accumulation is likely to change across different historical regimes of accumulation. Future research should focus on endogenizing θ by modeling it as a dynamic, time-varying parameter that responds to shifts in the choice of technique, distributive conflict, and institutional structures.

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A Derivation of Accumulation Dynamics under Unbalanced Growth

This appendix provides the derivation of the dynamical equations summarized in §3.3 and §3.4. All results follow from three components: the accounting identity $\hat{k} + \delta = \phi^p B$, the unbalanced growth closure $g_{Y^p} = \theta g_K$, and the investment-savings closure $\phi^p = s$.

A.1 The Unbalanced Growth Closure: Derivation of the Bernoulli ODE

The accounting identity under the investment-savings closure is:

$$\hat{k} + \delta = s B. \quad (24)$$

Differentiating with respect to time yields:

$$\frac{d\hat{k}}{dt} = s \dot{B}. \quad (25)$$

Expressing $\dot{B}$ in terms of the growth rate of capital productivity ($\hat{b}$):

$$\dot{B} = B \hat{b}. \quad (26)$$

Substituting this into equation (25) and utilizing the accounting identity $sB = \hat{k} + \delta$ results in the acceleration identity:

$$\frac{d\hat{k}}{dt} = (\hat{k} + \delta) \hat{b}. \quad (27)$$

Recall that capital productivity is $B \equiv Y^p/K$, which in growth rates implies $\hat{b} = \hat{y}^p - \hat{k}$. Under the unbalanced growth closure $g_{Y^p} = \theta g_K$ (or $\hat{y}^p = \theta \hat{k}$ in growth rate notation), this gives the growth rate of capital productivity as $\hat{b} = (\theta - 1) \hat{k}$. Substituting this into equation (27) provides the acceleration dynamic:

$$\boxed{\frac{d\hat{k}}{dt} = (\theta - 1) \hat{k} (\hat{k} + \delta)}, \quad \text{where } \hat{k} \text{ and } \delta \text{ are expressed in consistent temporal and physical units.} \quad (28)$$

This is a Bernoulli equation of order 2 in $\hat{k}$, where the nonlinearity arises from the product $\hat{k}(\hat{k} + \delta)$.

A.2 Analytical Solution

Expanding equation (28) gives:

$$\frac{d\hat{k}}{dt} = (\theta - 1) \hat{k}^2 + (\theta - 1) \delta \hat{k}. \quad (29)$$

Applying the Bernoulli substitution $v \equiv \hat{k}^{-1}$ with $\dot{v} = -\hat{k}^{-2} \dot{\hat{k}}$ transforms the equation into a first-order linear ODE:

$$\dot{v} + (\theta - 1) \delta v = -(\theta - 1). \quad (30)$$

The solution for $v(t)$, given initial conditions $v_0 = \hat{k}_0^{-1}$, is:

$$v(t) = \left(v_0 + \frac{1}{\delta} \right) e^{-(\theta-1)\delta t} - \frac{1}{\delta}. \quad (31)$$

Inverting $v(t)$ yields the explicit path for the net accumulation rate:

$$\hat{k}(t) = \frac{1}{\left(\frac{1}{\hat{k}_0} + \frac{1}{\delta} \right) e^{-(\theta-1)\delta t} - \frac{1}{\delta}}. \quad (32)$$

A.3 Equilibria and Stability Analysis

The equilibria $\hat{k}^*$ are defined by $(\theta - 1)\hat{k}(\hat{k} + \delta) = 0$. For $\theta \neq 1$, the roots are $\hat{k}_1^* = 0$ (stationary net capital) and $\hat{k}_2^* = -\delta$ (zero gross investment). Linearizing around $\hat{k}^* = 0$ via the Jacobian $f'(\hat{k}) = (\theta - 1)(2\hat{k} + \delta)$ yields:

- Over-accumulation ($\theta < 1$): $f'(0) < 0$ — the stagnation equilibrium is locally stable.
- Under-accumulation ($\theta > 1$): $f'(0) > 0$ — the stagnation equilibrium is unstable.

A.4 The Trend-Stabilized Closure: Derivation of the Quadratic ODE

Under the trend-stabilized closure $\hat{y}^p = \theta\hat{k} + b$, the growth rate of capital productivity is:

$$\hat{b} = (\theta - 1)\hat{k} + b. \quad (33)$$

Substituting this into the acceleration identity (equation 27) generates a quadratic ODE:

$$\frac{d\hat{k}}{dt} = (\hat{k} + \delta) \left[(\theta - 1)\hat{k} + b \right]. \quad (34)$$

The equilibria are the roots of $(\hat{k} + \delta)[(\theta - 1)\hat{k} + b] = 0$:

$$\hat{k}_1^* = -\delta, \quad \hat{k}_2^* = \frac{b}{1 - \theta}. \quad (35)$$

The Jacobian $g'(\hat{k}) = 2(\theta - 1)\hat{k} + (\theta - 1)\delta + b$ evaluated at $\hat{k}_2^*$ confirms local stability for the empirically relevant case ($\theta < 1$ and $b > 0$): the interior root $b/(1 - \theta)$ acts as a strictly positive attractor, preventing stagnation. As $b \rightarrow 0$, the trend-stabilized closure collapses to the unbalanced growth attractor at $\hat{k}^* = 0$.

B Data Construction and Diagnostic Support

B.1 Series Glossary

Table 12 lists the core replication variables, their sources, and transformation rules. All series derive from Shaikh’s (2016) data companions via Real Economic Analysis.⁴

Table 12: Replication series: symbols, sources, and transformations

Symbol / Code	Variable	Source (table, line)	Transformation
GVAcorp_t	Corporate gross value added (corrected)	NIPA 1.14 (L1) + imputed-interest adj. (Table 7.11)	nominal
KGCcorp_t	Corporate gross capital stock (GPIM-adjusted)	Fixed Asset 6.1 (L9); IRS Statistics of Income; GPIM recursion	nominal
KNCcorp_t	Corporate net capital stock (GPIM-adjusted)	same as above	nominal
KNCcorpbea_t	Corporate net capital stock (BEA 2011 official)	Fixed Asset 6.1 (L9)	nominal
p_t	Common deflator	pKN index, 2005=100	$\text{KNC}/\text{KNR} \times 100$
y_t	Real log output	constructed	$\ln(\text{VAcorp}_t/p_t)$
k_t	Real log capital	constructed	$\ln(\text{KGCcorp}_t/p_t)$
Profshcorp_t	Corporate profit share (corrected)	NIPA 1.14 (L8, L1–L2) + imputed-interest adj.	ratio

Note: All BEA and NIPA data downloaded in the replication vintage. Shaikh’s original series used data downloaded in 2011 (Shaikh, 2016, Appendix 6.7, fn. 1); discrepancies attributable to subsequent BEA revisions are possible and noted where material.

Source: [Real Economic Analysis Data Tables](#)

B.2 Imputed-Interest Correction

The NIPA corporate value added figure includes imputed financial intermediation services as a cost to non-financial corporations. The corporate imputed-interest adjustment removes this component:

$$\text{CorpImpIntAdj}_t \equiv -\text{BankMonIntPaid}_t - \text{CorpNFNetImpIntPaid}_t, \quad (36)$$

where the first term is NIPA Table 7.11 (lines 4+44+73 minus 28+52+91) and the second is Table 7.11 (line 74 minus 53). In 2009 the adjustment raises corporate net operating surplus by approximately 10%. Figure 8 displays the corrected versus uncorrected series and their ratio.

B.3 Corrected Corporate Output: GVAcorp_t

$$\text{GVAcorp}_t \equiv \text{GVAcorpnipa}_t + \text{CorpImpIntAdj}_t, \quad (37)$$

where GVAcorpnipa_t is NIPA Table 1.14 (line 1) and CorpImpIntAdj_t is defined in equation (36).

⁴Replication data and construction scripts are available at Real Economic Analysis: <https://www.realecon.org/data>.

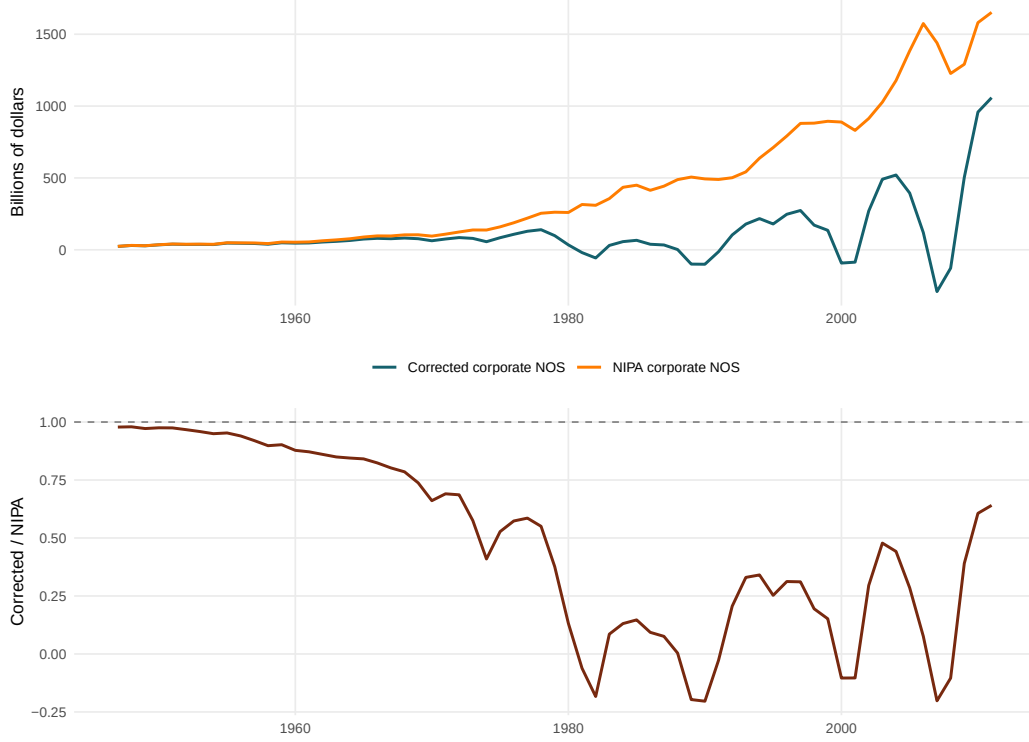


Figure 8: Imputed interest adjustment impact: corrected corporate NOS versus NIPA corporate NOS (1947–2011). Ratio shows magnitude of adjustment.

B.4 Reconstructed Corporate Capital Stock: $KGC_{corp,t}$

GPIM accumulation rule. The current-cost capital stock is accumulated as:

$$K_t = IG_t + z_t^* \cdot K_{t-1}, \quad z_t^* \equiv (1 - z_t) \frac{p_t^K}{p_{t-1}^K}, \quad (38)$$

where IG_t is nominal gross investment, z_t is the period depreciation rate, and z_t^* is the survival-and-revaluation factor that reflates the surviving stock to current replacement cost.

Three adjustments (applied before the recursion).

1. **Depletion rates.** BEA 1993 finite service lives replace the BEA 2011 infinite-lives assumption, implying higher depreciation and lower retirement rates consistent with physical exhaustion.
2. **Initial values.** BEA 1993 initial values anchored to 1925 replace the BEA 2011 starting values. The 1993 values are lower, producing faster accumulation growth and convergence to the official path by approximately 1977.
3. **Great Depression correction.** The IRS corporate book value index ([of the Census, 1975](#), Series V 115, pp. 924–926) is applied over 1925–1947, rescaling the BEA historical stock by the IRS/BEA ratio each year. The GPIM recursion resumes from 1947. Net effect: $KNC_{corp,t}$ is approximately 28 % lower in 1947 than the BEA 2011 official measure.

Inventories. IRS corporate balance sheet inventory ratios are extrapolated over 1947–1989 via a linear fit to the 1990–2011 IRS/BEA ratio, rescaled to the adjusted GPIM stocks, and added to $KGCcorp_t$. Source: IRS Statistics of Income 1926–2011; FRB Flow of Funds FL105015205.A for non-financial corporate inventories in current cost.

Accuracy. The average ratio of the GPIM approximation to the BEA current-cost stock over 1947–2005 is 99.60 %.

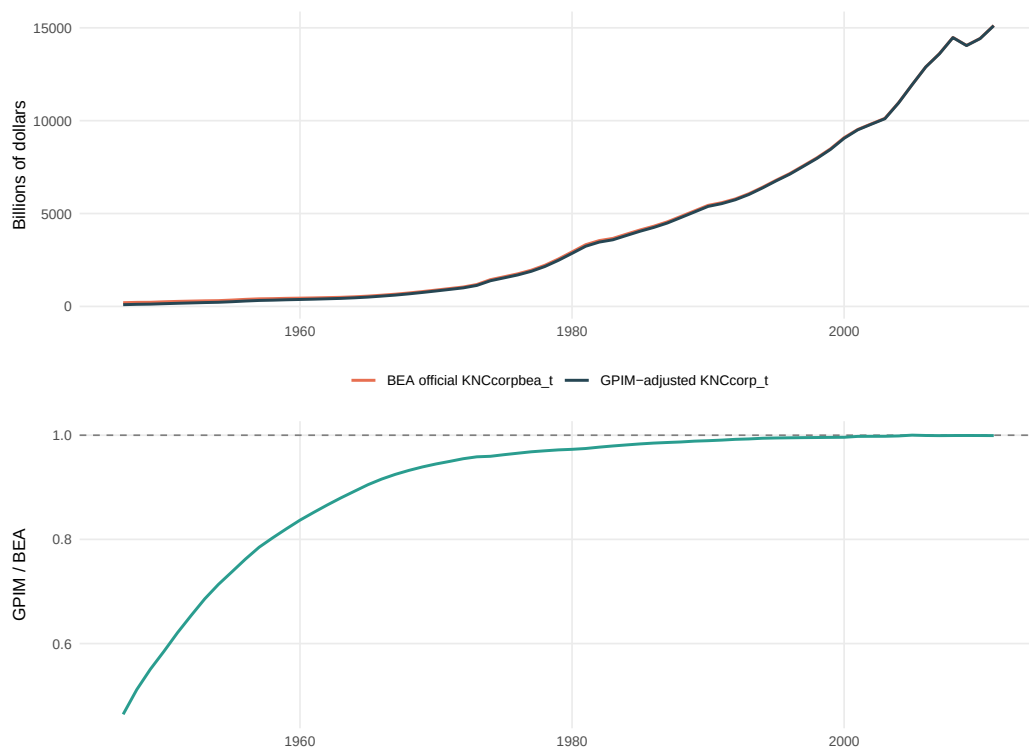


Figure 9: GPIM-adjusted net capital stock versus BEA 2011 official series (1947–2011). Ratio plot shows convergence by 1977.

B.5 Common Deflator: p_t

The replication uses the pKN implicit price deflator based at $100 = 2005$, estimated under GPIM rules. Current BEA releases employ quality-adjusted chain weights that adjust for quality vintage, breaking stock-flow consistency for accumulation accounting. The pKN index preserves stock-flow consistency, ensuring the output–capital relation reflects real productive dynamics rather than quality-adjustment artifacts (Shaikh, 2016, Appendix 6.6, Sec. II).

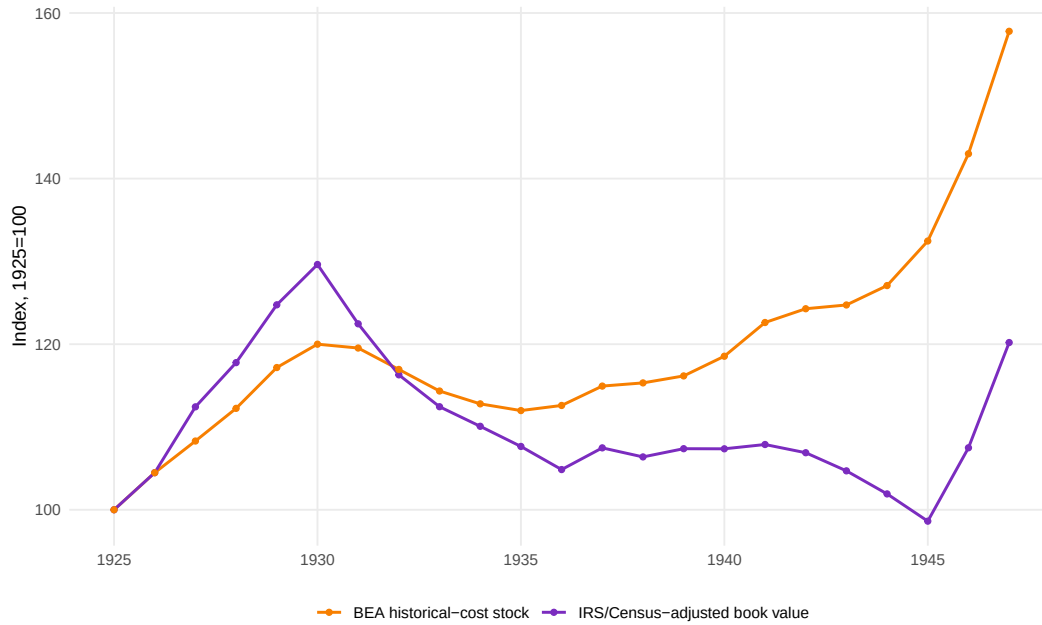


Figure 10: IRS corporate book value index versus BEA historical stock (1925–1947). Both indexed to 100 in 1925.

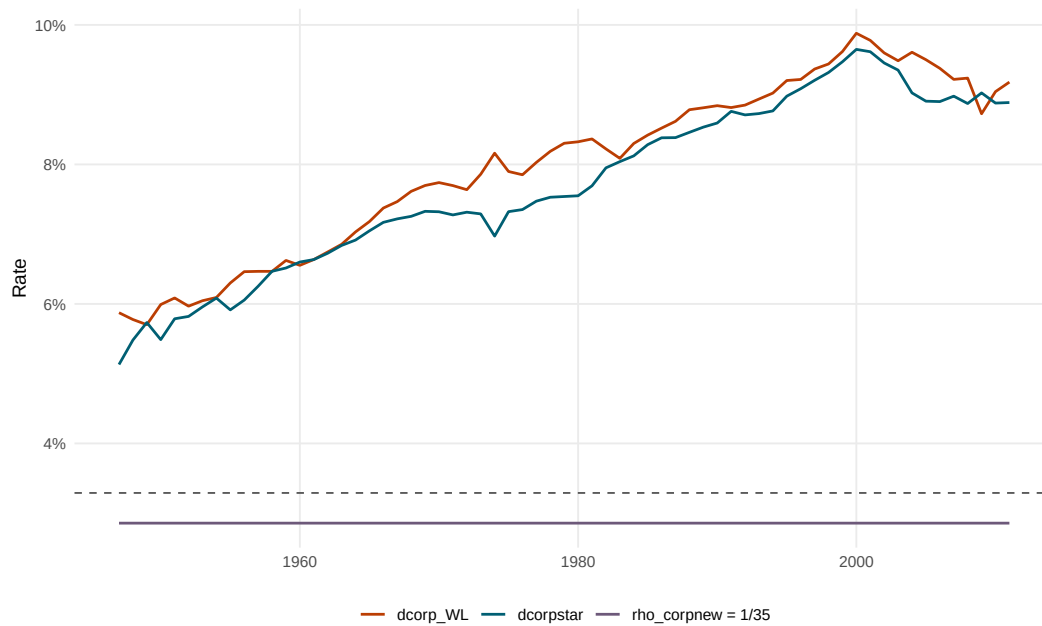


Figure 11: Depreciation and retirement rates under BEA 1993 finite service lives. Horizontal line marks critical value (3.29%).

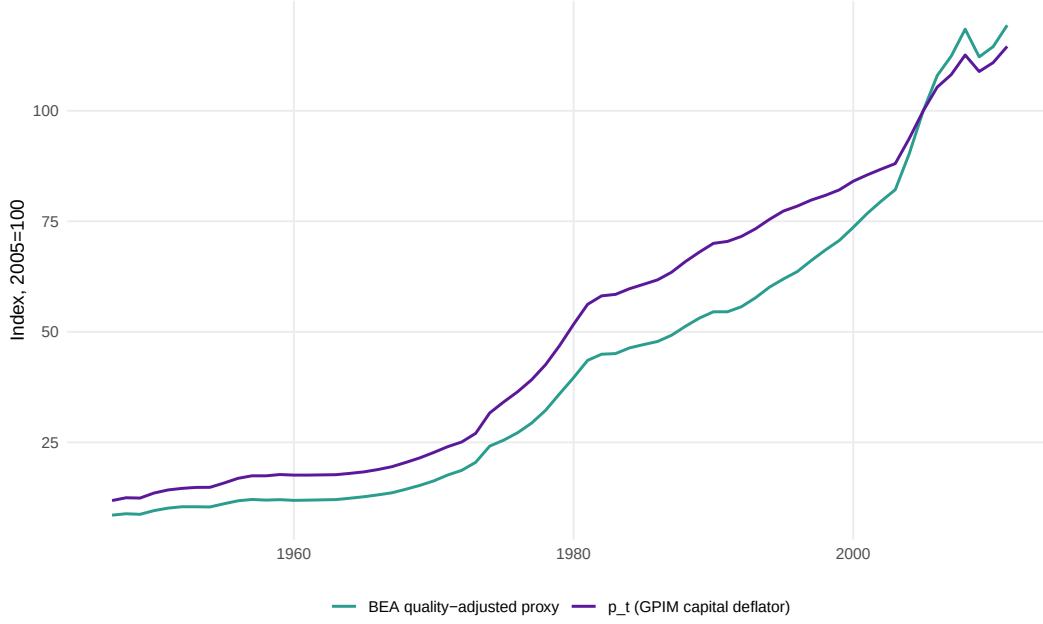


Figure 12: Common deflator index: GPIM-consistent pKN versus BEA quality-adjusted proxy (2005=100).

B.6 Corrected Corporate Profit Share: Profshcorp_t

The same imputed-interest adjustment applied to GVAcorp_t is applied to both the numerator and denominator of the profit share:

$$\text{NOScorp}_t \equiv \text{NOScorpnipa}_t + \text{CorpImpIntAdj}_t, \quad (39)$$

$$\text{VAcorp}_t \equiv \text{VAcorpnipa}_t + \text{CorpImpIntAdj}_t, \quad (40)$$

$$\text{Profshcorp}_t \equiv \frac{\text{NOScorp}_t}{\text{VAcorp}_t}, \quad (41)$$

where NOScorpnipa_t is NIPA Table 1.14 (line 8) and VAcorpnipa_t is NIPA Table 1.14 (line 1 minus line 2).

B.7 Descriptive Statistics and Unit Root Tests

Table 13 reports summary statistics for the core series. Table 14 presents unit root test results (ADF, Phillips-Perron) for y_t and k_t in levels and first differences.

Table 13: Core series summary statistics (1947–2011)

Variable	N	Mean	Median	Std Dev	Min	Max	Skewness	Kurtosis
y_t	65	14.9281	14.8653	0.5674	13.8866	15.7628	−0.1438	1.8398
k_t	65	15.9658	15.9810	0.7313	14.7979	17.1127	−0.0206	1.6872
Δy_t	64	0.0293	0.0355	0.0391	−0.0979	0.1055	−0.6331	3.4760
Δk_t	64	0.0362	0.0361	0.0078	0.0186	0.0530	−0.1127	2.4022

p_t	65	50.3486	46.8900	32.8159	11.8262	114.5539	0.3955	1.8063
D_{56}	65	0.0154	0.0000	0.1240	0.0000	1.0000	7.8750	63.0156
D_{74}	65	0.0154	0.0000	0.1240	0.0000	1.0000	7.8750	63.0156
D_{80}	65	0.0154	0.0000	0.1240	0.0000	1.0000	7.8750	63.0156

Note: Source: Real Economic Analysis (<https://www.realecon.org/data>), Shaikh (2016) data companions. Variables constructed per Appendix 12. Dummy variables are impulse indicators with mean $1/65 \approx 0.0154$.

Table 14: Unit root tests for core series (1947–2011)

Variable	Form	Test	Specification	Statistic	Lag/BW	Critical values			Reject at 5%
						1%	5%	10%	
y_t	level	ADF	none	3.0027	1	-2.60	-1.95	-1.61	no
y_t	level	PP	intercept	-1.3280	3	-3.53	-2.91	-2.59	no
y_t	level	PP	trend	-2.1341	3	-4.11	-3.48	-3.17	no
y_t	Δ	ADF	intercept	-5.0966	1	-3.51	-2.89	-2.58	yes
y_t	Δ	PP	intercept	-5.9637	3	-3.54	-2.91	-2.59	yes
y_t	Δ	PP	trend	-5.9460	3	-4.11	-3.48	-3.17	yes
k_t	level	ADF	none	1.1552	4	-2.60	-1.95	-1.61	no
k_t	level	PP	intercept	-0.3863	3	-3.53	-2.91	-2.59	no
k_t	level	PP	trend	-1.2114	3	-4.11	-3.48	-3.17	no
k_t	Δ	ADF	intercept	-2.0670	3	-3.51	-2.89	-2.58	no
k_t	Δ	PP	intercept	-1.9027	3	-3.54	-2.91	-2.59	no
k_t	Δ	PP	trend	-1.8431	3	-4.11	-3.48	-3.17	no

Note: ADF: Augmented Dickey–Fuller test with max lag 6 and AIC selection. PP: Phillips–Perron test with Newey–West bandwidth. "none" = no deterministic terms; "intercept" = constant only; "trend" = constant + linear trend. PP "none" specification not computed. Rejection at 5% indicates stationarity under that specification.

B.8 ARDL Specification Search and Bounds Testing

Table 15 displays the full specification grid ($p, q = 1-6$; Cases I–V) with information criteria, diagnostic tests, and recovered long-run coefficients. Asterisks mark case-specific AIC/BIC minima.

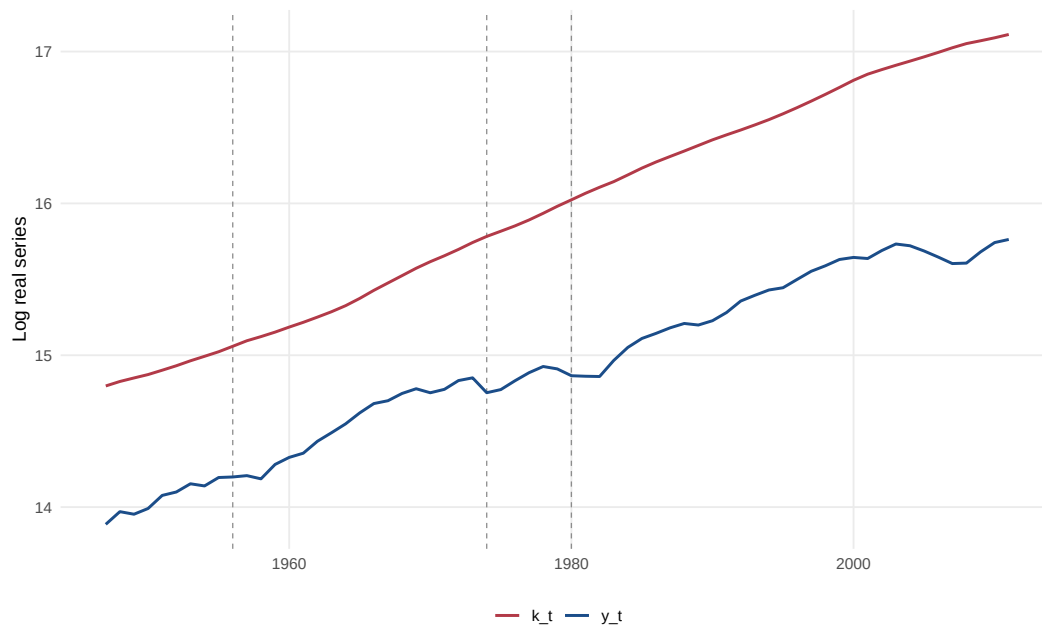


Figure 13: Log output (y_t) and log capital (k_t), 1947–2011. Vertical dashed lines mark 1956, 1974, and 1980.



Figure 14: Output-capital ratio ($y_t - k_t$), 1947–2011. Horizontal line marks long-run mean.

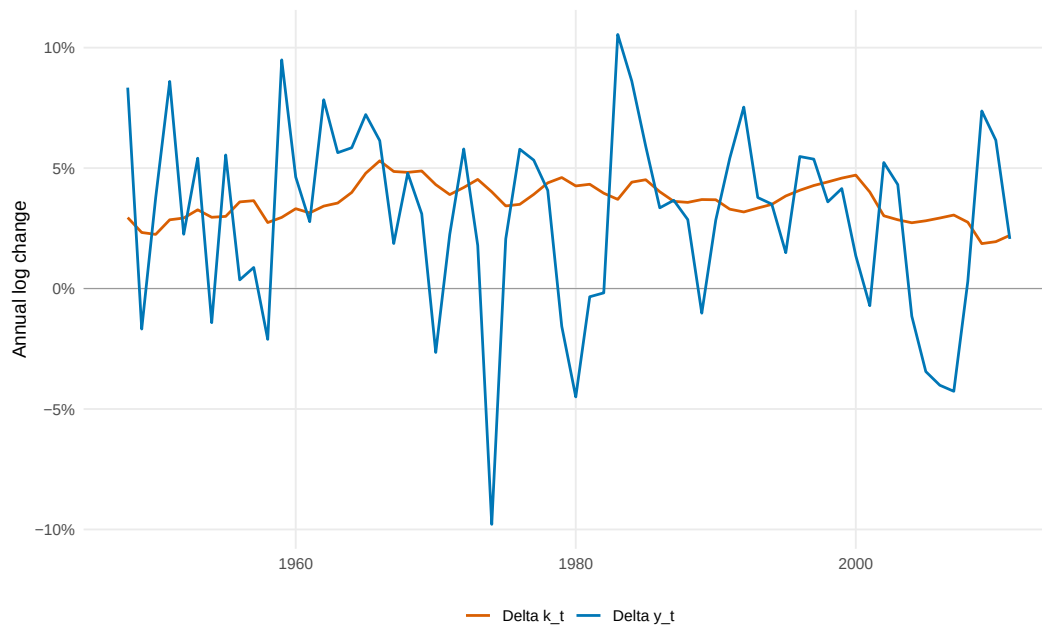


Figure 15: Annual growth rates of output (Δy_t) and capital (Δk_t), 1947–2011.

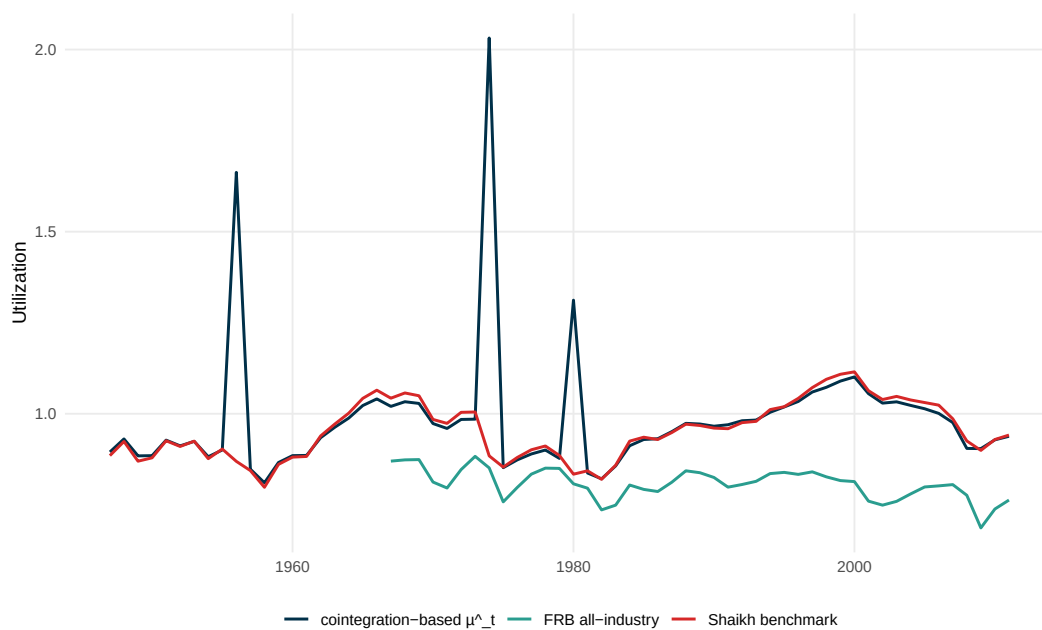


Figure 16: Capacity utilization comparison: Shaikh (cointegration-based) versus FRB survey measure, 1947–2011. Descriptive comparison only.

Table 15: ARDL specification search: information criteria and diagnostics (1947–2011)

p	q	Case	AIC	SBC	T_{eff}	$\hat{\theta}$	LM(1)	p -val	LM(4)	p -val	Best
1	1	I	−237.63	−222.52	64	0.8529	3.2749	0.070	5.0893	0.278	
1	1	II	−243.09	−225.82	64	0.6989	3.7150	0.054	6.9003	0.141	
1	1	III	−243.09	−225.82	64	0.6989	3.7150	0.054	6.9003	0.141	
1	1	IV	−243.09	−223.66	64	−0.5048	3.4486	0.063	6.5385	0.162	
1	1	V	−243.09	−223.66	64	−0.5048	3.4486	0.063	6.5385	0.162	
1	2	I	−239.33	−222.19	63	0.7273	7.0198	0.008	11.1845	0.025	
1	2	II	−242.69	−223.41	63	0.7110	7.1117	0.008	11.8692	0.018	
1	2	III	−242.69	−223.41	63	0.7110	7.1117	0.008	11.8692	0.018	
1	2	IV	−241.90	−220.47	63	−0.4718	7.0457	0.008	11.2449	0.024	
1	2	V	−241.90	−220.47	63	−0.4718	7.0457	0.008	11.2449	0.024	
1	3	I	−244.02	−224.88	62	0.8967	11.3499	0.001	14.1599	0.007	
1	3	II	−249.33	−228.05	62	0.6695	10.3605	0.001	12.7048	0.013	
1	3	III	−249.33	−228.05	62	0.6695	10.3605	0.001	12.7048	0.013	
1	3	IV	−249.62	−226.22	62	−0.8749	10.3560	0.001	11.9612	0.018	*
1	3	V	−249.62	−226.22	62	−0.8749	10.3560	0.001	11.9612	0.018	*
2	3	I	−250.91	−229.63	62	0.9301	5.4621	0.019	9.3503	0.053	
2	3	II	−262.79	−239.40	62	0.7023	0.5641	0.453	3.5633	0.468	*
2	3	III	−262.79	−239.40	62	0.7023	0.5641	0.453	3.5633	0.468	*
2	3	IV	−264.33	−238.81	62	−0.4836	0.2849	0.594	2.6102	0.625	*
2	3	V	−264.33	−238.81	62	−0.4836	0.2849	0.594	2.6102	0.625	*
4	5	I	−253.33	−224.01	60	0.7899	0.0461	0.830	7.6617	0.105	*
4	5	II	−257.75	−226.33	60	0.7102	0.2628	0.608	6.6363	0.156	
4	5	III	−257.75	−226.33	60	0.7102	0.2628	0.608	6.6363	0.156	
4	5	IV	−257.11	−223.60	60	−0.4117	0.2120	0.645	6.5141	0.164	
4	5	V	−257.11	−223.60	60	−0.4117	0.2120	0.645	6.5141	0.164	

Note: ARDL(p, q) specifications estimated with impulse dummies D_{56}, D_{74}, D_{80} . AIC/SBC: Akaike/Schwarz information criteria (lower = better). T_{eff} : effective sample size after lag construction. $\hat{\theta}$: long-run output–capital coefficient recovered from ECM parameters. LM(1)/LM(4): Breusch–Godfrey serial correlation tests (null = no autocorrelation). Asterisk (*) marks best specification by AIC/SBC within each deterministic case. Full grid ($p, q = 1$ –6; Cases I–V) available in replication code.

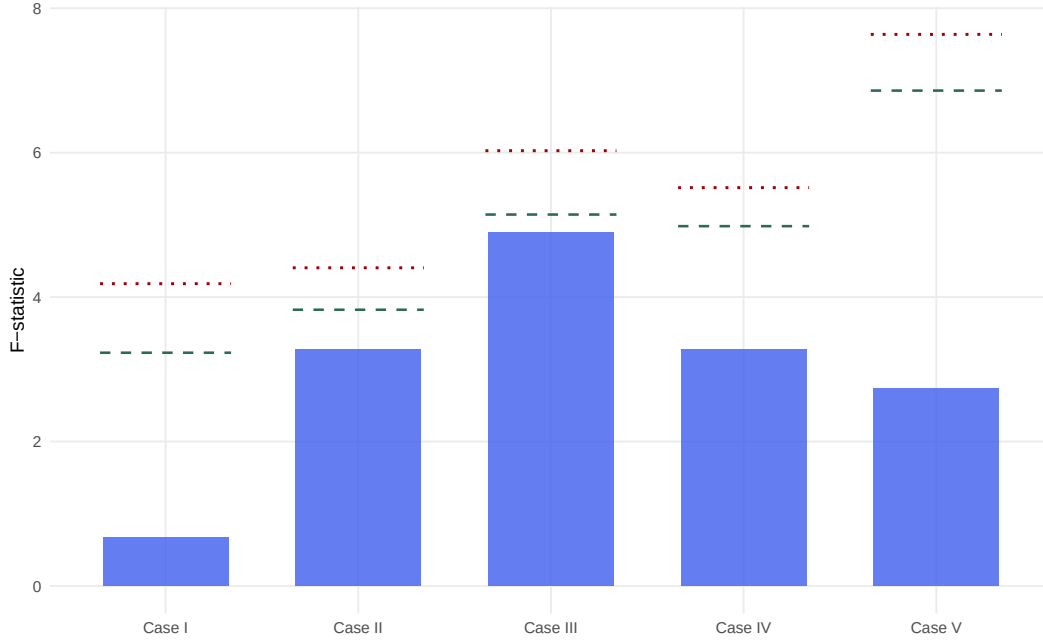


Figure 17: F -statistic across deterministic Cases I–V. Horizontal lines mark lower ($I(0)$) and upper ($I(1)$) critical bounds at 5% significance.

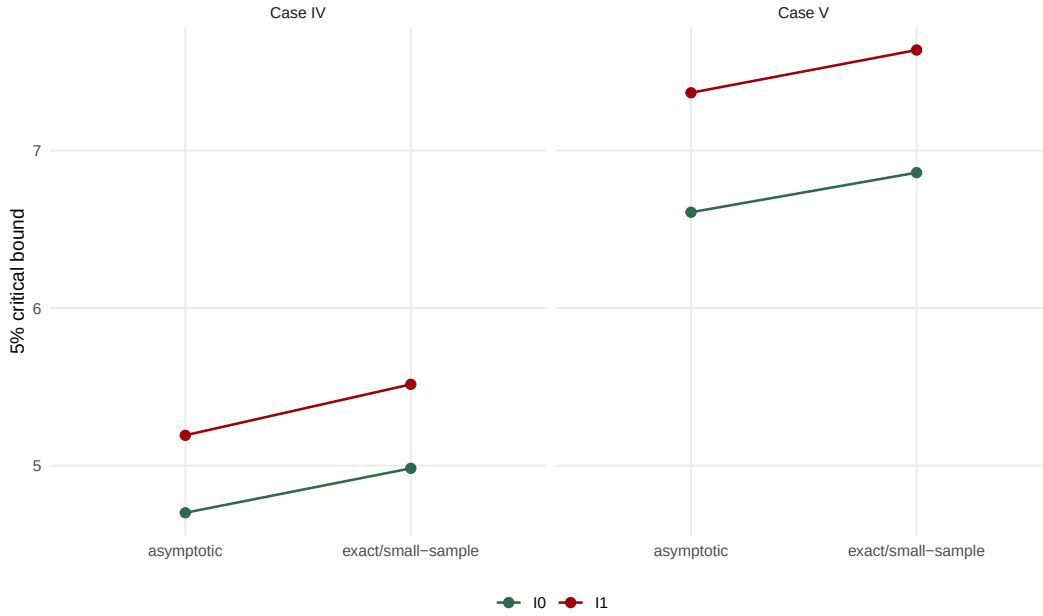


Figure 18: Exact-sample versus asymptotic critical bounds for Cases IV and V ($T = 65$). Exact bounds computed via stochastic simulation (Natsiopoulos and Tzeremes, 2022).

B.9 Dummy Variable Diagnostics

Figure 21 displays squared residuals from the baseline ARDL specification with impulse dummies D_{56} , D_{74} , D_{80} overlaid. The dummies absorb outlier-driven residual spikes identified via squared-error diagnostics (Patterson, 2000).

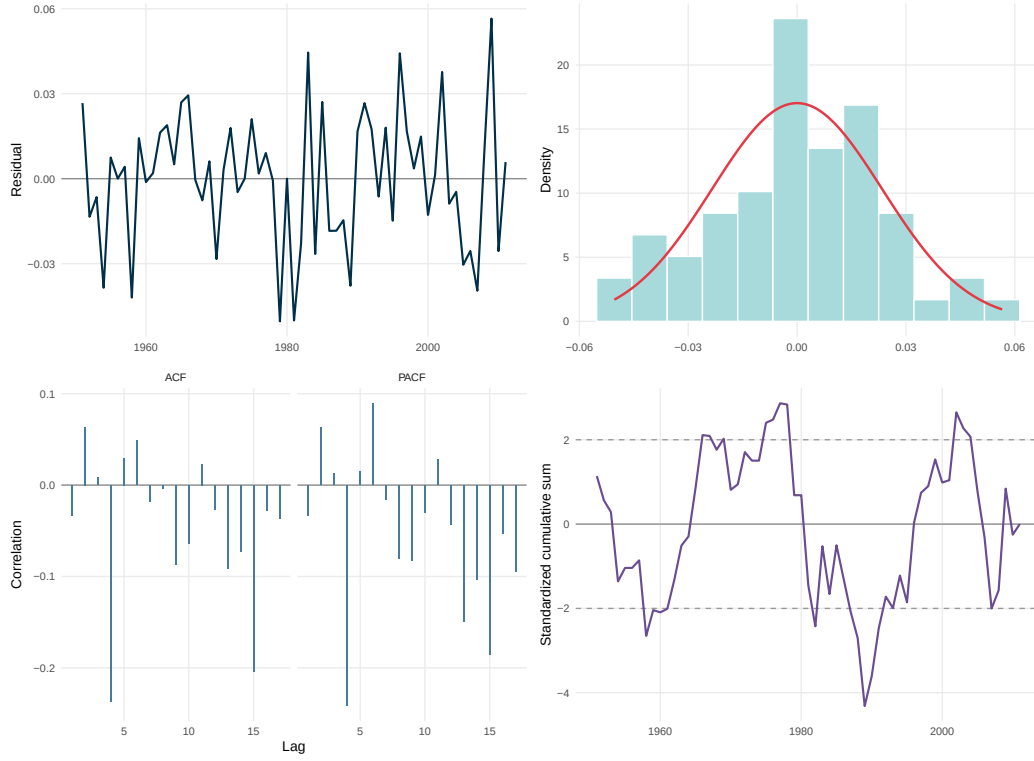


Figure 19: Residual diagnostics from baseline ARDL(2,4): (a) residuals over time, (b) density with normal curve, (c) ACF/PACF, (d) CUSUM test.

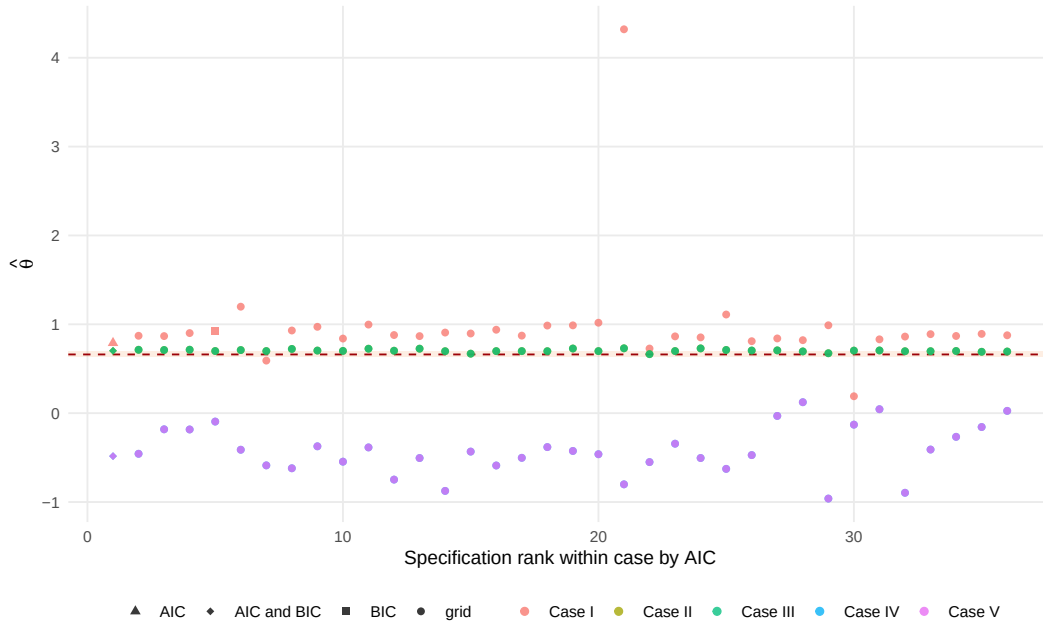


Figure 20: Long-run coefficient $\hat{\theta}$ estimates ranked by AIC across specification grid. Horizontal band marks Shaikh's $0.6609 \pm 5\%$.

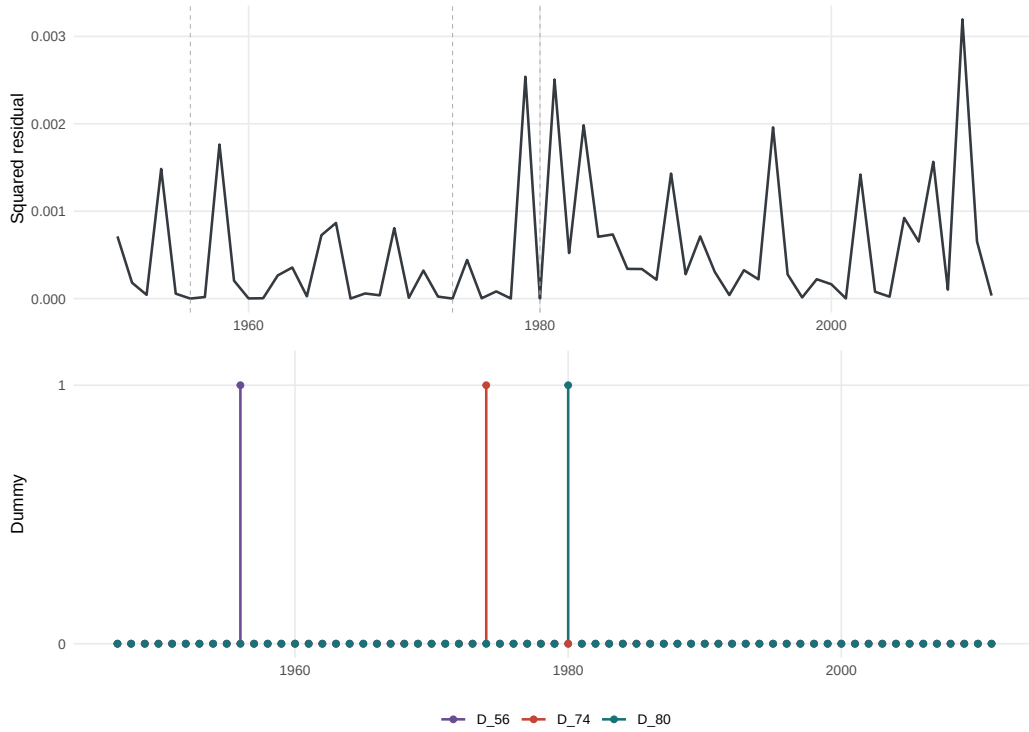


Figure 21: Squared residuals from baseline ARDL with impulse dummies D_{56} , D_{74} , D_{80} (1947–2011). Dummies absorb outlier-driven spikes.

B.10 Series Construction Protocol

Figure 22 visualizes the pipeline from raw data to fitted capacity utilization ($\hat{\mu}_t$). The flowchart traces corrections, deflation, GPIM recursion, ARDL estimation, and residualization.

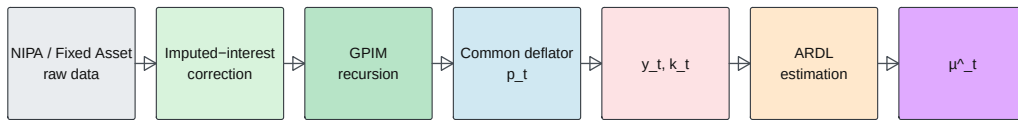


Figure 22: Series construction protocol: raw data $\rightarrow$ corrected output/GPIM capital $\rightarrow$ common deflator $\rightarrow$ ARDL estimation $\rightarrow \hat{\mu}_t$.